

SECOND EDITION

Digital Signal Processing

SYSTEM ANALYSIS AND DESIGN



Paulo S. R. Diniz
Eduardo A. B. da Silva
Sergio L. Netto

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Digital Signal Processing

This new, fully revised edition covers all the major topics of digital signal processing (DSP) design and analysis in a single, all-inclusive volume, interweaving theory with real-world examples and design trade-offs.

Building on the success of the original, this edition includes new material on random signal processing, a new chapter on spectral estimation, greatly expanded coverage of filter banks and wavelets, and new material on the solution of difference equations. Additional steps in mathematical derivations make them easier to follow, and an important new feature is the Do-it-Yourself section at the end of each chapter, where readers get hands-on experience of solving practical signal processing problems in a range of MATLAB[®] experiments.

With 120 worked examples, 20 case studies, and almost 400 homework exercises, the book is essential reading for anyone taking digital signal processing courses. Its unique blend of theory and real-world practical examples also makes it an ideal reference for practitioners.

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Second Edition

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and
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To our families,
our parents,
and our students.

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Preface

This book originated from a training course for engineers at the research and development center of TELEBRAS, the former Brazilian telecommunications holding. That course was taught by the first author back in 1987, and its main goal was to present efficient digital filter design methods suitable for solving some of their engineering problems. Later on, this original text was used by the first author as the basic reference for the digital filters and digital signal processing courses of the Electrical Engineering Program at COPPE/Federal University of Rio de Janeiro.

For many years, former students asked why the original text was not transformed into a book, as it presented a very distinct view that they considered worth publishing. Among the numerous reasons not to attempt such task, we could mention that there were already a good number of well-written texts on the subject; also, after many years of teaching and researching on this topic, it seemed more interesting to follow other paths than the painful one of writing a book; finally, the original text was written in Portuguese and a mere translation of it into English would be a very tedious task.

In later years, the second and third authors, who had attended the signal processing courses using the original material, were continuously giving new ideas on how to proceed. That was when we decided to go through the task of completing and updating the original text, turning it into a modern textbook. The book then took on its first-edition form, updating the original text, and including a large amount of new material written for other courses taught by the three authors up to 2002.

This second edition barely resembles the original lecture notes for several reasons. The original material was heavily concentrated on filter design and realization, whereas the present version includes a large amount of material on discrete-time systems, discrete transforms, spectral estimation, multirate systems, filter banks, and wavelets.

This book is mainly written for use as a textbook on a digital signal processing course for undergraduate students who have had previous exposure to basic linear systems, or to serve as a textbook on a graduate-level course where the most advanced topics of some chapters are covered. This reflects the structure we have at the Federal University of Rio de Janeiro, as well as at a number of other universities we have contact with. The second edition has a special feature designed for readers to test their learning by hands-on experience through so-called Do-it-yourself sections, with the aid of MATLAB[®]. A Do-it-yourself section is included in all chapters of the book. The book also includes, at the end of most chapters, a brief section aimed at giving a start to the reader on how to use MATLAB as a tool for the analysis and design of digital signal processing systems. As in the first edition, we decided that having explanations about MATLAB inserted in the main text would in some cases distract the readers, making them lose focus on the subject.

A distinctive feature of this book is to present a wide range of topics in digital signal processing design and analysis in a concise and complete form, while allowing the reader to fully develop practical systems. Although this book is primarily intended as an undergraduate and graduate textbook, its origins on training courses for industry warrant its potential usefulness to engineers working in the development of signal processing systems. In fact, our objective is to equip the readers with the tools that enable them to understand why and how to use digital signal processing systems; to show them how to approximate a desired transfer function characteristic using polynomials and ratios of polynomials; to teach them why an appropriate mapping of a transfer function into a suitable structure is important for practical applications; and to show how to analyze, represent, and explore the trade-off between the time and frequency representations of deterministic and stochastic signals. For all that, each chapter includes a number of examples and end-of-chapter problems to be solved. These are aimed at assimilating the concepts, as well as complementing the text. In particular, the second edition includes many new examples and exercises to be solved.

Chapters 1 and 2 review the basic concepts of discrete-time signal processing and z transforms. Although many readers may be familiar with these subjects, they could benefit from reading these chapters, getting used to the notation and the authors' way of presenting the subject. In Chapter 1 we review the concepts of discrete-time systems, including the representation of discrete-time signals and systems, as well as their time-domain responses. Most important, we present the sampling theorem, which sets the conditions for the discrete-time systems to solve practical problems related to our real continuous-time world. The basic concepts of random signals are also introduced in this chapter, followed by the Do-it-yourself section aiding the reader to test their progress in discrete-time signals and systems. Chapter 2 is concerned with the z and Fourier transforms, which are useful mathematical tools for representation of discrete-time signals and systems. The basic properties of the z and Fourier transforms are discussed, including a stability test in the z transform domain. The chapter also shows how the analysis of random signals can benefit from the z -domain formulation.

Chapter 3 discusses discrete transforms, with special emphasis given to the discrete Fourier transform (DFT), which is an invaluable tool in the frequency analysis of discrete-time signals. The DFT allows a discrete representation of discrete-time signals in the frequency domain. Since the sequence representation is natural for digital computers, the DFT is a very powerful tool, because it enables us to manipulate frequency-domain information in the same way as we can manipulate the original sequences. The importance of the DFT is further increased by the fact that computationally efficient algorithms, the so-called fast Fourier transforms (FFTs), are available to compute the DFT. This chapter also presents real coefficient transforms, such as cosine and sine transforms, which are widely used in modern audio and video coding, as well as in a number of other applications. A discussion about orthogonality in transforms is also included. This section also includes a discussion on the several forms of representing the signals, in order to aid the reader with the available choices.

Chapter 4 addresses the basic structures for mapping a transfer function into a digital filter. It is also devoted to some basic analysis methods and properties of digital filter structures.

The chapter also introduces some simple and useful building blocks widely utilized in some designs and applications.

Chapter 5 introduces several approximation methods for filters with finite-duration impulse response (FIR), starting with the simpler frequency sampling method and the widely used windows method. This method also provides insight to the windowing strategy used in several signal processing applications. Other approximation methods included are the maximally flat filters and those based on the weighted least-squares (WLS) method. This chapter also presents the Chebyshev approximation based on a multivariable optimization algorithm called the Remez exchange method. This approach leads to linear-phase transfer functions with minimum order given a prescribed set of frequency response specifications. This chapter also discusses the WLS–Chebyshev method which leads to transfer functions where the maximum and the total energy of the approximation error are prescribed. This approximation method is not widely discussed in the open literature but appears to be very useful for a number of applications.

Chapter 6 discusses the approximation procedures for filters with infinite-duration impulse response (IIR). We start with the classical continuous-time transfer-function approximations, namely the Butterworth, Chebyshev, and elliptic approximations, that can generate discrete-time transfer functions by using appropriate transformations. Two transformation methods are then presented: the impulse-invariance and the bilinear transformation methods. The chapter also includes a section on frequency transformations in the discrete-time domain. The simultaneous magnitude and phase approximation of IIR digital filters using optimization techniques is also included, providing a tool to design transfer functions satisfying more general specifications. The chapter closes by addressing the issue of time-domain approximations.

Chapter 7 introduces the basic concepts of classical estimation theory. It starts by describing the nonparametric spectral estimation methods based on a periodogram, followed by the minimum-variance spectral estimator. The chapter continues with a discussion on modeling theory, addressing the rational transfer function models and presenting the Yule–Walker equations. Several parametric spectral estimation methods are also presented, namely: the linear prediction method; the covariance method; the autocorrelation method; the Levinson–Durbin algorithm; and Burg’s method. The chapter also discusses the Wiener filter as an extension of the linear prediction method.

Chapter 8 deals with basic principles of discrete-time systems with multiple sampling rates. In this chapter we emphasize the basic properties of multirate systems, thoroughly addressing the decimation and interpolation operations, giving examples of their use for efficient digital filter design. The chapter discusses many key properties of multirate systems, such as inverse operations and noble identities, and introduces some analytical tools, such as polyphase decomposition and the commutator models. In addition, we discuss the concepts of overlapped block filtering, which can be very useful in some fast implementations of digital signal processing building blocks. The chapter also includes some discussion on how decimators and interpolators affect the properties of random signals.

Chapter 9 discusses some properties pertaining to the internal structure of filter banks, followed by the concept and construction of perfect reconstruction filter banks. The chapter also

includes some analysis tools and classifications for the filter banks and transmultiplexers. This chapter presents several design techniques for multirate filter banks, including several forms of two-band filter banks, cosine-modulated filter banks, and lapped transforms.

Chapter 10 introduces the concepts of time–frequency analysis and the discrete wavelet transform. It also presents the multiresolution representation of signals through wavelet transforms and discusses the design of wavelet transforms using filter banks. In addition, some design techniques to generate orthogonal (as well as biorthogonal) bases for signal representation are presented. Several properties of wavelets required for their classification, design, and implementation are discussed in this chapter.

Chapter 11 provides a brief introduction to the binary number representations most widely used in the implementation of digital signal processing systems. The chapter also explains how the basic elements utilized in these systems work and discusses a particular, and yet instructive, type of implementation based on distributed arithmetic. Chapter 11 also includes the models that account for quantization effects in digital filters. We discuss several approaches to analyze and deal with the effects of representing signals and filter coefficients with finite wordlength. In particular, we study the effects of quantization noise in products, signal scaling that limits the internal signal dynamic range, coefficient quantization in the designed transfer function, and the nonlinear oscillations which may occur in recursive realizations. These analyses are used to indicate the filter realizations that lead to practical finite-precision implementations of digital filters.

In Chapter 12 we present some techniques to reduce the computational complexity of FIR filters with demanding specifications or specialized requirements. The first structure discussed is the lattice form, which finds application in a number of areas, including the design of filter banks. Several useful implementation forms of FIR filters, such as polyphase, frequency-domain, recursive running sum, and modified-sinc forms, are presented to be employed as building blocks in several design methods. In particular, we introduce the prefilter and interpolation methods which are mainly useful in designing narrowband lowpass and highpass filters. In addition, we present the frequency-response masking approach, for designing filters with narrow transition bands satisfying more general specifications, and the quadrature method, for narrow bandpass and bandstop filters.

Chapter 13 presents a number of efficient realizations for IIR filters. For these filters, a number of realizations considered efficient from the finite-precision effects point of view are presented and their salient features are discussed in detail. These realizations will equip the reader with a number of choices for the design of good IIR filters. Several families of structures are considered in this chapter, namely: parallel and cascade designs using direct-form second-order sections; parallel and cascade designs using section-optimal and limit-cycle-free state-space sections; lattice filters; and several forms of wave digital filters. In addition, this chapter includes a discussion on doubly complementary filters and their use in the implementation of quadrature mirror filter banks.

This book contains enough material for an undergraduate course on digital signal processing and a first-year graduate course. There are many alternative ways to compose these courses; in the following we describe some recommendations that have been employed successfully in signal processing courses.

- An undergraduate course in discrete-time systems or digital signal processing at junior level. This should include most parts of Chapters 1, 2, 3, 4, and the nonparametric methods of Chapter 7. It could also include the noniterative approximation methods of Chapters 5 and 6, namely the frequency sampling and window methods described in Chapter 5, the analog-based approximation methods, and also the continuous-time to discrete-time transformation methods for IIR filtering of Chapter 6.
- An undergraduate course in digital signal processing at senior level. This should briefly review parts of Chapters 1 and 2 and cover Chapters 3, 4, and 7. It could also include the noniterative approximation methods of Chapters 5 and 6, namely the frequency sampling and window methods described in Chapter 5, the analog-based approximation methods, and also the continuous-time to discrete-time transformation methods for IIR filtering of Chapter 6. Chapters 8 and 11 could complement the course.
- An undergraduate course in digital filtering at senior level or first-year graduate. This should cover Chapter 4 and the iterative approximation methods of Chapters 5 and 6. The course could also cover selected topics from Chapters 11, 12, and 13. At the instructor's discretion, the course could also include selected parts of Chapter 8.
- As a graduate course textbook on multirate systems, filter banks and wavelets. The course could cover Chapters 8, 9, and 10, as well as the lattice form in Chapter 12 and doubly complementary filters in Chapter 13.

Obviously, there are several other choices for courses based on the material of this book which will depend on the course length and the judicious choice of the instructor.

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We sincerely hope that the book reflects the harmony, pleasure, friendship, and tenderness that we experience working together. Our partnership was written in the stars and heaven sent.

Introduction

When we hear the word “signal” we may first think of a phenomenon that occurs continuously over time that carries some information. Most phenomena observed in nature are continuous in time, such as for instance our speech or heart beating, the temperature of the city where we live, the car speed during a given trip, the altitude of the airplane we are traveling in – these are typical continuous-time signals. Engineers are always devising ways to design systems, which are in principle continuous time, for measuring and interfering with these and other phenomena.

One should note that, although continuous-time signals pervade our daily lives, there are also several signals which are originally discrete time; for example, the stock-market weekly financial indicators, the maximum and minimum daily temperatures in our cities, and the average lap speed of a racing car.

If an electrical or computer engineer has the task of designing systems to interact with natural phenomena, their first impulse is to convert some quantities from nature into electric signals through a transducer. Electric signals, which are represented by voltages or currents, have a continuous-time representation. Since digital technology constitutes an extremely powerful tool for information processing, it is natural to think of processing the electric signals generated using it. However, continuous-time signals cannot be processed using computer technology (digital machines), which are especially suited to deal with sequential computation involving numbers. Fortunately, this fact does not prevent the use of digital integrated circuits (which is the technology behind the computer technology revolution we witness today) in signal processing systems designs. This is because many signals taken from nature can be fully represented by their sampled versions, where the sampled signals coincide with the original continuous-time signals at some instants in time. If we know how fast the important information changes, then we can always sample and convert continuous-time information into discrete-time information which, in turn, can be converted into a sequence of numbers and transferred to a digital machine.

The main advantages of digital systems relative to analog systems are high reliability, suitability for modifying the system’s characteristics, and low cost. These advantages motivated the digital implementation of many signal processing systems which used to be implemented with analog circuit technology. In addition, a number of new applications became viable once the very large scale integration (VLSI) technology was available. Usually in the VLSI implementation of a digital signal processing system the concern is to reduce power consumption and/or area, or to increase the circuit’s speed in order to meet the demands of high-throughput applications.

Currently, a single digital integrated circuit may contain millions of logic gates operating at very high speeds, allowing very fast digital processors to be built at a reasonable cost. This technological development opened avenues to the introduction of powerful general-purpose computers which can be used to design and implement digital signal processing systems. In addition, it allowed the design of microprocessors with special features for signal processing, namely the digital signal processors (DSPs). As a consequence, there are several tools available to implement very complex digital signal processing systems. In practice, a digital signal processing system is implemented either by software on a general-purpose digital computer or DSP, or by using application-specific hardware, usually in the form of an integrated circuit.

For the reasons explained above, the field of digital signal processing has developed so fast in recent decades that it has been incorporated into the graduate and undergraduate programs of virtually all universities. This is confirmed by the number of good textbooks available in this area: Oppenheim & Schaffer (1975, 1989); Rabiner & Gold (1975); Peled & Liu (1985); Roberts & Mullis (1987); Ifeachor & Jervis (1993); Jackson (1996); Antoniou (2006); Mitra (2006); Proakis & Manolakis (2007). The present book is aimed at equipping readers with tools that will enable them to design and analyze most digital signal processing systems. The building blocks for digital signal processing systems considered here are used to process signals which are discrete in time and in amplitude. The main tools emphasized in this text are:

- discrete-time signal representations
- discrete transforms and their fast algorithms
- spectral estimation
- design and implementation of digital filters and digital signal processing systems
- multirate systems and filter banks
- wavelets.

Transforms and filters are the main parts of linear signal processing systems. Although the techniques we deal with are directly applicable to processing deterministic signals, many statistical signal processing methods employ similar building blocks in some way, as will be clear in the text.

Digital signal processing is extremely useful in many areas. In the following, we enumerate a few of the disciplines where the topics covered by this book have found application.

- (a) *Image processing*: An image is essentially a space-domain signal; that is, it represents a variation of light intensity and color in space. Therefore, in order to process an image using an analog system, it has to be converted into a time-domain signal, using some form of scanning process. However, to process an image digitally, there is no need to perform this conversion, for it can be processed directly in the spatial domain, as a matrix of numbers. This lends the digital image processing techniques enormous power. In fact, in image processing, two-dimensional signal representation, filtering, and transforms play a central role (Jain, 1989).

- (b) *Multimedia systems*: Such systems deal with different kinds of information sources, such as image, video, audio, and data. In such systems, the information is essentially represented in digital form. Therefore, it is crucial to remove redundancy from the information, allowing compression, and thus efficient transmission and storage (Jayant & Noll, 1984; Gersho & Gray, 1992; Bhaskaran & Konstantinides, 1997). The Internet is a good application example where information files are transferred in a compressed manner. Most of the compression standards for images use transforms and quantizers.

The transforms, filter banks, and wavelets are very popular in compression applications because they are able to exploit the high correlation of signal sources such as audio, still images, and video (Malvar, 1992; Fliege, 1994; Vetterli and Kovačević, 1995; Strang and Nguyen, 1996; Mallat, 1999).

- (c) *Communication systems*: In communication systems, the compression and coding of the information sources are also crucial, since services can be provided at higher speed or to more users by reducing the amount of data to be transmitted. In addition, channel coding, which consists of inserting redundancy in the signal to compensate for possible channel distortions, may also use special types of digital filtering (Stüber, 1996).

Communication systems usually include fixed filters, as well as some self-designing filters for equalization and channel modeling which fall in the class of adaptive filtering systems (Diniz, 2008). Although these filters employ a statistical signal processing framework (Hayes, 1996; Kay, 1988; Manolakis *et al.*, 2000) to determine how their parameters should change, they also use some of the filter structures and in some cases the transforms introduced in this book.

Many filtering concepts take part on modern multiuser communication systems employing code-division multiple access (Verdu, 1998).

Wavelets, transforms, and filter banks also play a crucial role in the conception of orthogonal frequency-division multiplexing (OFDM) (Akansu & Medley, 1999), which is used in digital audio and TV broadcasting.

- (d) *Audio signal processing*: In statistical signal processing the filters are designed based on observed signals, which might imply that we are estimating the parameters of the model governing these signals (Kailath *et al.*, 2000). Such estimation techniques can be employed in digital audio restoration (Godsill & Rayner, 1998), where the resulting models can be used to restore lost information. However, these estimation models can be simplified and made more effective if we use some kind of sub-band processing with filter banks and transforms (Kahrs & Brandenburg, 1998). In the same field, digital filters were found to be suitable for reverberation algorithms and as models for musical instruments (Kahrs & Brandenburg, 1998).

In addition to the above applications, digital signal processing is at the heart of modern developments in speech analysis and synthesis, mobile radio, sonar, radar, biomedical engineering, seismology, home appliances, and instrumentation, among others. These developments occurred in parallel with the advances in the technology of transmission,

processing, recording, reproduction, and general treatment of signals through analog and digital electronics, as well as other means such as acoustics, mechanics, and optics.

We expect that, with the digital signal processing tools described in this book, the reader will be able to proceed further, not only exploring and understanding some of the applications described above, but developing new ones as well.

1.1 Introduction

Digital signal processing is the discipline that studies the rules governing the behavior of discrete signals, as well as the systems used to process them. It also deals with the issues involved in processing continuous signals using digital techniques. Digital signal processing pervades modern life. It has applications in compact disc players, computer tomography, geological processing, mobile phones, electronic toys, and many others.

In analog signal processing, we take a continuous signal, representing a continuously varying physical quantity, and pass it through a system that modifies this signal for a certain purpose. This modification is, in general, continuously variable by nature; that is, it can be described by differential equations.

Alternatively, in digital signal processing, we process sequences of numbers using some sort of digital hardware. We usually call these sequences of numbers digital or discrete-time signals. The power of digital signal processing comes from the fact that, once a sequence of numbers is available to appropriate digital hardware, we can carry out any form of numerical processing on it. For example, suppose we need to perform the following operation on a continuous-time signal:

$$y(t) = \frac{\cosh \left[\ln(|x(t)|) + x^3(t) + \cos^3 \left(\sqrt{|x(t)|} \right) \right]}{5x^5(t) + e^{x(t)} + \tan(x(t))}. \quad (1.1)$$

This would be clearly very difficult to implement using analog hardware. However, if we sample the analog signal $x(t)$ and convert it into a sequence of numbers $x(n)$, it can be input to a digital computer, which can perform the above operation easily and reliably, generating a sequence of numbers $y(n)$. If the continuous-time signal $y(t)$ can be recovered from $y(n)$, then the desired processing has been successfully performed.

This simple example highlights two important points. One is how powerful digital signal processing is. The other is that, if we want to process an analog signal using this sort of resource, we must have a way of converting a continuous-time signal into a discrete-time one, such that the continuous-time signal can be recovered from the discrete-time signal. However, it is important to note that very often discrete-time signals do not come from continuous-time signals, that is, they are originally discrete-time, and the results of their processing are only needed in digital form.

In this chapter, we study the basic concepts of the theory of discrete-time signals and systems. We emphasize the treatment of discrete-time systems as separate entities from

continuous-time systems. We first define discrete-time signals and, based on this, we define discrete-time systems, highlighting the properties of an important subset of these systems, namely linearity and time invariance, as well as their description by discrete-time convolutions. We then study the time-domain response of discrete-time systems by characterizing them using difference equations. We close the chapter with Nyquist's sampling theorem, which tells us how to generate, from a continuous-time signal, a discrete-time signal from which the continuous-time signal can be completely recovered. Nyquist's sampling theorem forms the basis of the digital processing of continuous-time signals.

1.2 Discrete-time signals

A discrete-time signal is one that can be represented by a sequence of numbers. For example, the sequence

$$\{x(n), \quad n \in \mathbb{Z}\}, \quad (1.2)$$

where $\mathbb{Z}$ is the set of integer numbers, can represent a discrete-time signal where each number $x(n)$ corresponds to the amplitude of the signal at an instant nT . If $x_a(t)$ is an analog signal, we have that

$$x(n) = x_a(nT), \quad n \in \mathbb{Z}. \quad (1.3)$$

Since n is an integer, T represents the interval between two consecutive points at which the signal is defined. It is important to note that T is not necessarily a time unit. For example, if $x_a(t)$ is the temperature along a metal rod, then if T is a length unit, $x(n) = x_a(nT)$ may represent the temperature at sensors placed uniformly along this rod.

In this text, we usually represent a discrete-time signal using the notation in Equation (1.2), where $x(n)$ is referred to as the n th sample of the signal (or the n th element of the sequence). An alternative notation, used in many texts, is to represent the signal as

$$\{x_a(nT), \quad n \in \mathbb{Z}\}, \quad (1.4)$$

where the discrete-time signal is represented explicitly as samples of an analog signal $x_a(t)$. In this case, the time interval between samples is explicitly shown; that is, $x_a(nT)$ is the sample at time nT . Note that, using the notation in Equation (1.2), a discrete-time signal whose adjacent samples are 0.03 s apart would be represented as

$$\dots \quad x(0), x(1), x(2), x(3), x(4), \dots, \quad (1.5)$$

whereas, using Equation (1.4) it would be represented as

$$\dots \quad x_a(0), x_a(0.03), x_a(0.06), x_a(0.09), x_a(0.12), \dots \quad (1.6)$$

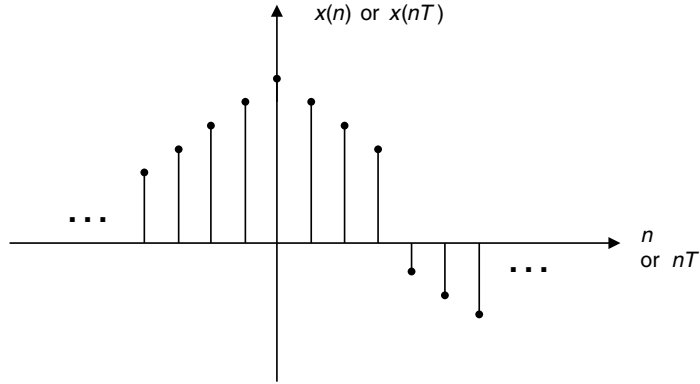


Fig. 1.1.

General representation of a discrete-time signal.

The graphical representation of a general discrete-time signal is shown in Figure 1.1. In what follows, we describe some of the most important discrete-time signals.

Unit impulse (see Figure 1.2a):

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0. \end{cases} \quad (1.7)$$

Delayed unit impulse (see Figure 1.2b):

$$\delta(n - m) = \begin{cases} 1, & n = m \\ 0, & n \neq m. \end{cases} \quad (1.8)$$

Unit step (see Figure 1.2c):

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0. \end{cases} \quad (1.9)$$

Cosine function (see Figure 1.2d):

$$x(n) = \cos(\omega n). \quad (1.10)$$

The angular frequency of this sinusoid is ω rad/sample and its frequency is $\omega/2\pi$ cycles/sample. For example, in Figure 1.2d, the cosine function has angular frequency $\omega = 2\pi/16$ rad/sample. This means that it completes one cycle, that equals 2π radians, in 16 samples. If the sample separation represents time, then ω can be given in rad/(time unit). It is important to note that

$$\cos[(\omega + 2k\pi)n] = \cos(\omega n + 2kn\pi) = \cos(\omega n) \quad (1.11)$$

for $k \in \mathbb{Z}$. This implies that, in the case of discrete signals, there is an ambiguity in defining the frequency of a sinusoid. In other words, when referring to discrete sinusoids, ω and $\omega + 2k\pi$, $k \in \mathbb{Z}$, are the same frequency.

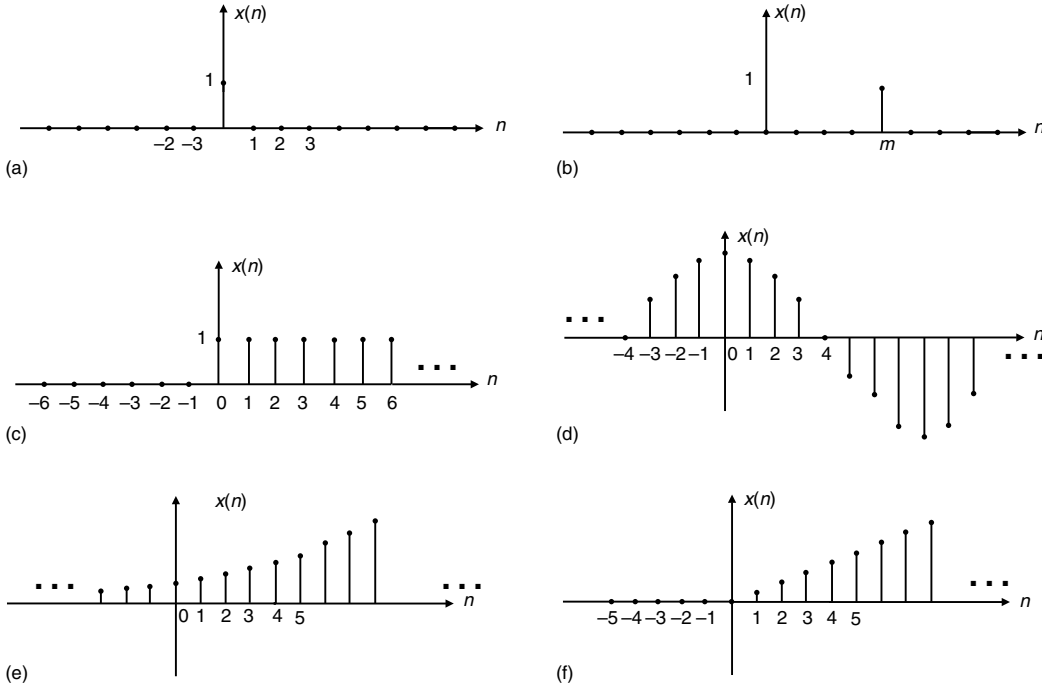


Fig. 1.2. Basic discrete-time functions: (a) unit impulse; (b) delayed unit impulse; (c) unit step; (d) cosine function with $\omega = 2\pi/16$ rad/sample; (e) real exponential function with $a = 0.2$; (f) unit ramp.

Real exponential function (see Figure 1.2e):

$$x(n) = e^{an}. \quad (1.12)$$

Unit ramp (see Figure 1.2f):

$$r(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (1.13)$$

By examining Figure 1.2b–f, we notice that any discrete-time signal is equivalent to a sum of shifted impulses multiplied by a constant; that is, the impulse shifted by k samples is multiplied by $x(k)$. This can also be deduced from the definition of a shifted impulse in Equation (1.8). For example, the unit step $u(n)$ in Equation (1.9) can also be expressed as

$$u(n) = \sum_{k=0}^{\infty} \delta(n - k). \quad (1.14)$$

Likewise, any discrete-time signal $x(n)$ can be expressed as

$$x(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n - k). \quad (1.15)$$

An important class of discrete-time signals or sequences is that of periodic sequences. A sequence $x(n)$ is periodic if and only if there is an integer $N \neq 0$ such that $x(n) = x(n + N)$ for all n . In such a case, N is called a period of the sequence. Note that, using this definition and referring to Equation (1.10), a period of the cosine function is an integer N such that

$$\cos(\omega n) = \cos[\omega(n + N)], \quad \text{for all } n \in \mathbb{Z}. \quad (1.16)$$

This happens only if there is $k \in \mathbb{N}$ such that $\omega N = 2\pi k$. The smallest period is then

$$N = \min_{\substack{k \in \mathbb{N} \\ (2\pi/\omega)k \in \mathbb{N}}} \left\{ \frac{2\pi}{\omega} k \right\}. \quad (1.17)$$

Therefore, we notice that not all discrete cosine sequences are periodic, as illustrated in Example 1.1. An example of a periodic cosine sequence with period $N = 16$ samples is given in Figure 1.2d.

Example 1.1. Determine whether the discrete signals above are periodic; if they are, determine their periods.

- (a) $x(n) = \cos[(12\pi/5)n]$
- (b) $x(n) = 10 \sin^2 \left[(7\pi/12)n + \sqrt{2} \right]$
- (c) $x(n) = 2 \cos(0.02n + 3)$.

Solution

- (a) In this case, we must have

$$\frac{12\pi}{5}(n + N) = \frac{12\pi}{5}n + 2k\pi \quad \Rightarrow \quad N = \frac{5k}{6}. \quad (1.18)$$

This implies that the smallest N results for $k = 6$. Then the sequence is periodic with period $N = 5$. Note that in this case

$$\cos\left(\frac{12\pi}{5}n\right) = \cos\left(\frac{2\pi}{5}n + 2\pi n\right) = \cos\left(\frac{2\pi}{5}n\right) \quad (1.19)$$

and thus we have also that the frequency of this sinusoid, besides being $\omega = 12\pi/5$, is also $\omega = 2\pi/5$, as indicated by Equation (1.11).

- (b) In this case, periodicity implies that

$$\sin^2 \left[\frac{7\pi}{12}(n + N) + \sqrt{2} \right] = \sin^2 \left(\frac{7\pi}{12}n + \sqrt{2} \right) \quad (1.20)$$

and then

$$\sin \left[\frac{7\pi}{12}(n + N) + \sqrt{2} \right] = \pm \sin \left(\frac{7\pi}{12}n + \sqrt{2} \right) \quad (1.21)$$

such that

$$\frac{7\pi}{12}(n+N) = \frac{7\pi}{12}n + k\pi \Rightarrow N = \frac{12k}{7}. \quad (1.22)$$

The smallest N results for $k = 7$. Then this discrete-time signal is periodic with period $N = 12$.

(c) The periodicity condition requires that

$$\cos[0.02(n+N) + 3] = \cos(0.02n + 3) \quad (1.23)$$

such that

$$0.02(n+N) = 0.02n + 2k\pi \Rightarrow N = 100k\pi. \quad (1.24)$$

Since no integer N satisfies the above equation, the sequence is not periodic.

△

1.3 Discrete-time systems

A discrete-time system maps an input sequence $x(n)$ to an output sequence $y(n)$, such that

$$y(n) = \mathcal{H}\{x(n)\}, \quad (1.25)$$

where the operator $\mathcal{H}\{\cdot\}$ represents a discrete-time system, as shown in Figure 1.3. Depending on the properties of $\mathcal{H}\{\cdot\}$, the discrete-time system can be classified in several ways, the most basic ones being either linear or nonlinear, either time invariant or time variant, and causal or noncausal. These classifications will be discussed in what follows.

1.3.1 Linearity

Let us suppose that there is a system that accepts as input a voice signal and outputs the voice signal modified such that its acute components (high frequencies) are enhanced. In such a system, it would be undesirable if, in the case that one increased the voice tone at the input, the output became distorted instead of enhanced. Actually, one tends to expect

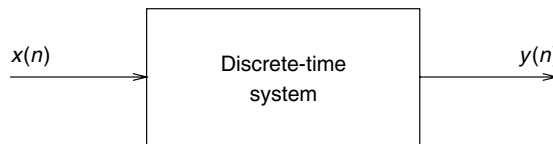


Fig. 1.3. Representation of a discrete-time system.

that if one speaks twice as loud at the input, then the output will be just twice as loud, with its acute components enhanced in the same way. Likewise, if two people speak at the same time, one would expect the system to enhance the high frequencies of both voices, each one in the same way as if they were individually input to the system. A system with such a behavior is what is referred to as a linear system. Such systems, besides being useful in many practical applications, have very nice mathematical properties. This fact makes linear systems an important class of discrete-time systems; therefore, they constitute the primary subject of this book.

In more precise terms, a discrete-time system is linear if and only if

$$\mathcal{H}\{ax(n)\} = a\mathcal{H}\{x(n)\} \quad (1.26)$$

and

$$\mathcal{H}\{x_1(n) + x_2(n)\} = \mathcal{H}\{x_1(n)\} + \mathcal{H}\{x_2(n)\} \quad (1.27)$$

for any constant a and any sequences $x(n)$, $x_1(n)$, and $x_2(n)$.

1.3.2 Time invariance

Sometimes it is desirable to have a system whose properties do not vary in time. In other words, one wants its input–output behavior to be the same irrespective of the time instant that the input is applied to the system. Such a system is referred to as a time-invariant system. As will be seen later, when combined with linearity, time invariance gives rise to an important family of systems.

In more precise terms, a discrete-time system is time invariant if and only if, for any input sequence $x(n)$ and integer n_0 , given that

$$\mathcal{H}\{x(n)\} = y(n) \quad (1.28)$$

then

$$\mathcal{H}\{x(n - n_0)\} = y(n - n_0). \quad (1.29)$$

Some texts refer to the time-invariance property as the shift-invariance property, since a discrete system can process samples of a function not necessarily in time, as emphasized before.

1.3.3 Causality

One of the main limitations of the time domain is that time always flows from past to present and, therefore, one cannot know the future. Although this statement might seem a bit too

philosophical, this concept has a strong influence on the way discrete-time systems can be used in practice. This is so because when processing a signal in time one cannot use future values of the input to compute the output at a given time. This leads to the definition of a causal system, that is a system that cannot “see into the future.”

In more precise terms, a discrete-time system is causal if and only if, when $x_1(n) = x_2(n)$ for $n < n_0$, then

$$\mathcal{H}\{x_1(n)\} = \mathcal{H}\{x_2(n)\}, \text{ for } n < n_0. \quad (1.30)$$

In other words, causality means that the output of a system at instant n does not depend on any input occurring after n .

It is important to note that, usually, in the case of a discrete-time signal, a noncausal system is not implementable in real time. This is because we would need input samples at instants of time greater than n in order to compute the output at time n . This would be allowed only if the time samples were pre-stored, as in off-line or batch implementations. It is important to note that if the signals to be processed do not consist of time samples acquired in real time, then there might be nothing equivalent to the concepts of past or future samples. Therefore, in these cases, the role of causality is of a lesser importance. For example, in Section 1.2 we mentioned a discrete signal that corresponds to the temperature at sensors uniformly spaced along a metal rod. For this discrete signal, a processor can have access to all its samples simultaneously. Therefore, in this case, even a noncausal system can be easily implemented.

Example 1.2. Characterize the following systems as being either linear or nonlinear, either time invariant or time varying, and causal or noncausal:

- (a) $y(n) = (n + b)x(n - 4)$
- (b) $y(n) = x^2(n + 1)$.

Solution

- (a) • Linearity:

$$\begin{aligned} \mathcal{H}\{ax(n)\} &= (n + b)ax(n - 4) \\ &= a(n + b)x(n - 4) \\ &= a\mathcal{H}\{x(n)\} \end{aligned} \quad (1.31)$$

and

$$\begin{aligned} \mathcal{H}\{x_1(n) + x_2(n)\} &= (n + b)[x_1(n - 4) + x_2(n - 4)] \\ &= (n + b)x_1(n - 4) + (n + b)x_2(n - 4) \\ &= \mathcal{H}\{x_1(n)\} + \mathcal{H}\{x_2(n)\}; \end{aligned} \quad (1.32)$$

therefore, the system is linear.

- Time invariance:

$$y(n - n_0) = [(n - n_0) + b]x[(n - n_0) - 4] \quad (1.33)$$

and then

$$\mathcal{H}\{x(n - n_0)\} = (n + b)x[(n - n_0) - 4] \quad (1.34)$$

such that $y(n - n_0) \neq \mathcal{H}\{x(n - n_0)\}$, and the system is time varying.

- Causality: if

$$x_1(n) = x_2(n), \quad \text{for } n < n_0 \quad (1.35)$$

then

$$x_1(n - 4) = x_2(n - 4), \quad \text{for } n - 4 < n_0 \quad (1.36)$$

such that

$$x_1(n - 4) = x_2(n - 4), \quad \text{for } n < n_0 \quad (1.37)$$

and then

$$(n + b)x_1(n - 4) = (n + b)x_2(n - 4), \quad \text{for } n < n_0. \quad (1.38)$$

Hence, $\mathcal{H}\{x_1(n)\} = \mathcal{H}\{x_2(n)\}$ for all $n < n_0$ and, consequently, the system is causal.

- (b) • Linearity:

$$\mathcal{H}\{ax(n)\} = a^2x^2(n + 1) \neq a\mathcal{H}\{x(n)\}; \quad (1.39)$$

therefore, the system is nonlinear.

- Time invariance:

$$\mathcal{H}\{x(n - n_0)\} = x^2[(n - n_0) + 1] = y(n - n_0), \quad (1.40)$$

so the system is time invariant.

- Causality:

$$\mathcal{H}\{x_1(n)\} = x_1^2(n + 1) \quad (1.41)$$

$$\mathcal{H}\{x_2(n)\} = x_2^2(n + 1). \quad (1.42)$$

Therefore, if $x_1(n) = x_2(n)$, for $n < n_0$, and $x_1(n_0) \neq x_2(n_0)$, then, for $n = n_0 - 1 < n_0$,

$$\mathcal{H}\{x_1(n_0 - 1)\} = x_1^2(n_0) \quad (1.43)$$

$$\mathcal{H}\{x_2(n_0 - 1)\} = x_2^2(n_0) \quad (1.44)$$

and we have that $\mathcal{H}\{x_1(n)\} \neq \mathcal{H}\{x_2(n)\}$ and the system is noncausal. $\triangle$

1.3.4 Impulse response and convolution sums

Suppose that $\mathcal{H}\{\cdot\}$ is a linear system and we apply an excitation $x(n)$ to the system. Since, from Equation (1.15), $x(n)$ can be expressed as a sum of shifted impulses

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k), \quad (1.45)$$

we can express its output as

$$y(n) = \mathcal{H}\left\{\sum_{k=-\infty}^{\infty} x(k)\delta(n-k)\right\} = \sum_{k=-\infty}^{\infty} \mathcal{H}\{x(k)\delta(n-k)\}. \quad (1.46)$$

Since $x(k)$ in the above equation is just a constant, the linearity of $\mathcal{H}\{\cdot\}$ also implies that

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\mathcal{H}\{\delta(n-k)\} = \sum_{k=-\infty}^{\infty} x(k)h_k(n), \quad (1.47)$$

where $h_k(n) = \mathcal{H}\{\delta(n-k)\}$ is the response of the system to an impulse at $n = k$.

If the system is also time invariant, and we define

$$\mathcal{H}\{\delta(n)\} = h_0(n) = h(n), \quad (1.48)$$

then $\mathcal{H}\{\delta(n-k)\} = h(n-k)$, and the expression in Equation (1.47) becomes

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k), \quad (1.49)$$

indicating that a linear time-invariant system is completely characterized by its unit impulse response $h(n)$. This is a very powerful and convenient result, which will be explored further, and lends great importance and usefulness to the class of linear time-invariant discrete-time systems. One should note that, when the system is linear and time varying, we would need, in order to compute $y(n)$, the values of $h_k(n)$, which depend on both n and k . This makes the computation of the summation in Equation (1.47) quite complex.

Equation (1.49) is called a convolution sum or a discrete-time convolution.¹ If we make the change of variables $l = n - k$, Equation (1.49) can be written as

$$y(n) = \sum_{l=-\infty}^{\infty} x(n-l)h(l); \quad (1.50)$$

¹ This operation is often also referred to as a discrete-time linear convolution in order to differentiate it from the discrete-time circular convolution, which will be defined in Chapter 3.

that is, we can interpret $y(n)$ as the result of the convolution of the excitation $x(n)$ with the system impulse response $h(n)$. A shorthand notation for the convolution operation, as described in Equations (1.49) and (1.50), is

$$y(n) = x(n) * h(n) = h(n) * x(n). \quad (1.51)$$

Suppose now that the output $y(n)$ of a system with impulse response $h(n)$ is the excitation for a system with impulse response $h'(n)$. In this case we have

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \quad (1.52)$$

$$y'(n) = \sum_{l=-\infty}^{\infty} y(l)h'(n-l). \quad (1.53)$$

Substituting Equation (1.52) in Equation (1.53), we have that

$$\begin{aligned} y'(n) &= \sum_{l=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x(k)h(l-k) \right) h'(n-l) \\ &= \sum_{k=-\infty}^{\infty} x(k) \left(\sum_{l=-\infty}^{\infty} h(l-k)h'(n-l) \right). \end{aligned} \quad (1.54)$$

By performing the change of variables $l = n - r$, the above equation becomes

$$\begin{aligned} y'(n) &= \sum_{k=-\infty}^{\infty} x(k) \left(\sum_{r=-\infty}^{\infty} h(n-r-k)h'(r) \right) \\ &= \sum_{k=-\infty}^{\infty} x(k) (h(n-k) * h'(n-k)) \\ &= \sum_{k=-\infty}^{\infty} x(n-k) (h(k) * h'(k)), \end{aligned} \quad (1.55)$$

showing that the impulse response of a linear time-invariant system formed by the series (cascade) connection of two linear time-invariant subsystems is the convolution of the impulse responses of the two subsystems.

Example 1.3. Compute $y(n)$ for the system depicted in Figure 1.4, as a function of the input signal and of the impulse responses of the subsystems.

Solution

From the previous results, it is easy to conclude that

$$y(n) = (h_2(n) + h_3(n)) * h_1(n) * x(n). \quad (1.56)$$

△

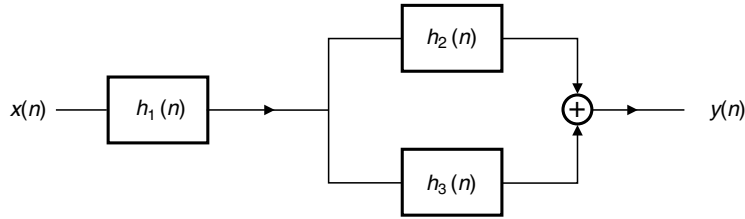


Fig. 1.4.

Linear time-invariant system composed of the connection of three subsystems.

1.3.5 Stability

A system is referred to as bounded-input bounded-output (BIBO) stable if, for every input limited in amplitude, the output signal is also limited in amplitude. For a linear time-invariant system, Equation (1.50) implies that

$$|y(n)| \leq \sum_{k=-\infty}^{\infty} |x(n-k)| |h(k)|. \quad (1.57)$$

The input being limited in amplitude is equivalent to

$$|x(n)| \leq x_{\max} < \infty, \text{ for all } n. \quad (1.58)$$

Therefore:

$$|y(n)| \leq x_{\max} \sum_{k=-\infty}^{\infty} |h(k)|. \quad (1.59)$$

Hence, we can conclude that a sufficient condition for a system to be BIBO stable is

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty, \quad (1.60)$$

since this condition forces $y(n)$ to be limited. To prove that this condition is also necessary, suppose that it does not hold; that is, that the summation in Equation (1.60) is infinite. If we choose an input such that

$$x(n_0 - k) = \begin{cases} +1, & \text{for } h(k) \geq 0 \\ -1, & \text{for } h(k) < 0, \end{cases} \quad (1.61)$$

we then have that

$$y(n_0) = |y(n_0)| = \sum_{k=-\infty}^{\infty} |h(k)|; \quad (1.62)$$

that is, the output $y(n)$ is unbounded, thus completing the necessity proof.

We can then conclude that the necessary and sufficient condition for a linear time-invariant system to be BIBO stable is given in Equation (1.60).

1.4 Difference equations and time-domain response

In most applications, discrete-time systems can be described by difference equations, which are the equivalent, for the discrete-time domain, to differential equations for the continuous-time domain. In fact, the systems that can be specified by difference equations are powerful enough to cover most practical applications. The input and output of a system described by a linear difference equation are generally related by (Gabel & Roberts, 1980)

$$\sum_{i=0}^N a_i y(n-i) - \sum_{l=0}^M b_l x(n-l) = 0. \quad (1.63)$$

This difference equation has an infinite number of solutions $y(n)$, like the solutions of differential equations in the continuous case. For example, suppose that a particular $y_p(n)$ satisfies Equation (1.63), that is

$$\sum_{i=0}^N a_i y_p(n-i) - \sum_{l=0}^M b_l x(n-l) = 0, \quad (1.64)$$

and that $y_h(n)$ is a solution to the homogeneous equation, that is

$$\sum_{i=0}^N a_i y_h(n-i) = 0. \quad (1.65)$$

Then, from Equations (1.63)–(1.65), we can easily infer that $y(n) = y_p(n) + y_h(n)$ is also a solution to the same difference equation.

The homogeneous solution $y_h(n)$ of a difference equation of order N , as in Equation (1.63), has N degrees of freedom (depends on N arbitrary constants). Therefore, one can only determine a solution for a difference equation if one supplies N auxiliary conditions. One example of a set of auxiliary conditions is given by the values of $y(-1), y(-2), \dots, y(-N)$. It is important to note that any N independent auxiliary conditions would be enough to solve a difference equation. In general, however, one commonly uses N consecutive samples of $y(n)$ as auxiliary conditions.

Example 1.4. Find the solution for the difference equation

$$y(n) = ay(n-1) \quad (1.66)$$

as a function of the initial condition $y(0)$.

Solution

Running the difference equation from $n = 1$ onwards, we have that

$$\left. \begin{aligned} y(1) &= ay(0) \\ y(2) &= ay(1) \\ y(3) &= ay(2) \\ \vdots \\ y(n) &= ay(n-1) \end{aligned} \right\}. \quad (1.67)$$

Multiplying the above equations, we have that

$$y(1)y(2)y(3) \dots y(n) = a^n y(0)y(1)y(2) \dots y(n-1); \quad (1.68)$$

therefore, the solution of the difference equation is

$$y(n) = a^n y(0). \quad (1.69)$$

△

Example 1.5. Solve the following difference equation:

$$y(n) = e^{-\beta} y(n-1) + \delta(n). \quad (1.70)$$

Solution

By making $a = e^{-\beta}$ and $y(0) = K$ in Example 1.4, we can deduce that any function of the form $y_h(n) = K e^{-\beta n}$ satisfies

$$y_h(n) = e^{-\beta} y_h(n-1) \quad (1.71)$$

and is therefore a solution of the homogeneous difference equation. Also, one can verify by substitution that $y_p(n) = e^{-\beta n} u(n)$ is a particular solution of Equation (1.70).

Therefore, the general solution of the difference equation is given by

$$y(n) = y_p(n) + y_h(n) = e^{-\beta n} u(n) + K e^{-\beta n}, \quad (1.72)$$

where the value of K is determined by the auxiliary conditions. Since this difference equation is of first order, we need to specify only one condition. For example, if we have that $y(-1) = \alpha$, the solution to Equation (1.70) becomes

$$y(n) = e^{-\beta n} u(n) + \alpha e^{-\beta(n+1)}. \quad (1.73)$$

△

Since a linear system must satisfy equation (1.26), it is clear that $\mathcal{H}\{0\} = 0$ for a linear system; that is, the output for a zero input must be zero. If we restrict ourselves to inputs that are null prior to a certain sample (that is, $x(n) = 0$, for $n < n_0$), then there is an interesting

relation between linearity, causality, and the initial conditions of a system. If the system is causal, then the output at $n < n_0$ cannot be influenced by any sample of the input $x(n)$ for $n \geq n_0$. Therefore, if $x(n) = 0$, for $n < n_0$, then $\mathcal{H}\{0\}$ and $\mathcal{H}\{x(n)\}$ must be identical for all $n < n_0$. Since, if the system is linear, $\mathcal{H}\{0\} = 0$, then necessarily $\mathcal{H}\{x(n)\} = 0$ for $n < n_0$. This is equivalent to saying that the auxiliary conditions for $n < n_0$ must be null. Such a system is referred to as being initially relaxed. Conversely, if the system is not initially relaxed, one cannot guarantee that it is causal. This will be made clearer in Example 1.6.

Example 1.6. For the linear system described by

$$y(n) = e^{-\beta}y(n-1) + u(n), \quad (1.74)$$

determine its output for the auxiliary conditions

- (a) $y(1) = 0$
- (b) $y(-1) = 0$

and discuss the causality in both situations.

Solution

The homogeneous solution of Equation (1.74) is the same as in Example 1.5; that is

$$y_h(n) = K e^{-\beta n}. \quad (1.75)$$

By direct substitution in Equation (1.74), it can be verified that the particular solution is of the form (see Section 1.5 for a method to determine such solutions)

$$y_p(n) = (a + b e^{-\beta n})u(n), \quad (1.76)$$

where

$$a = \frac{1}{1 - e^{-\beta}} \quad \text{and} \quad b = \frac{-e^{-\beta}}{1 - e^{-\beta}}. \quad (1.77)$$

Thus, the general solution of the difference equation is given by

$$y(n) = \left[\frac{1 - e^{-\beta(n+1)}}{1 - e^{-\beta}} \right] u(n) + K e^{-\beta n}. \quad (1.78)$$

(a) For the auxiliary condition $y(1) = 0$, we have that

$$y(1) = \frac{1 - e^{-2\beta}}{1 - e^{-\beta}} + K e^{-\beta} = 0, \quad (1.79)$$

yielding $K = -(1 + e^{\beta})$, and the general solution becomes

$$y(n) = \left[\frac{1 - e^{-\beta(n+1)}}{1 - e^{-\beta}} \right] u(n) - \left[e^{-\beta n} + e^{-\beta(n-1)} \right]. \quad (1.80)$$

Since for $n < 0$ we have that $u(n) = 0$, then $y(n)$ simplifies to

$$y(n) = -\left[e^{-\beta n} + e^{-\beta(n-1)}\right]. \quad (1.81)$$

Clearly, in this case, $y(n) \neq 0$, for $n < 0$, whereas the input $u(n) = 0$ for $n < 0$. Thus, the system is not initially relaxed and therefore is noncausal.

Another way of verifying that the system is noncausal is by noting that if the input is doubled, becoming $x(n) = 2u(n)$ instead of $u(n)$, then the particular solution is also doubled. Hence, the general solution of the difference equation becomes

$$y(n) = \left[\frac{2 - 2e^{-\beta(n+1)}}{1 - e^{-\beta}}\right]u(n) + Ke^{-\beta n}. \quad (1.82)$$

If we require that $y(1) = 0$, then $K = 2 + 2e^{\beta}$, and, for $n < 0$, this yields

$$y(n) = -2\left[e^{-\beta n} + e^{-\beta(n-1)}\right]. \quad (1.83)$$

Since this is different from the value of $y(n)$ for $u(n)$ as input, we see that the output for $n < 0$ depends on the input for $n > 0$; therefore, the system is noncausal.

(b) For the auxiliary condition $y(-1) = 0$, we have that $K = 0$, yielding the solution

$$y(n) = \left[\frac{1 - e^{-\beta(n+1)}}{1 - e^{-\beta}}\right]u(n). \quad (1.84)$$

In this case, $y(n) = 0$, for $n < 0$; that is, the system is initially relaxed and, therefore, causal, as discussed above. $\triangle$

Example 1.6 shows that the system described by the difference equation is noncausal because it has nonzero auxiliary conditions prior to the application of the input to the system. To guarantee both causality and linearity for the solution of a difference equation, we have to impose zero auxiliary conditions for the samples preceding the application of the excitation to the system. This is the same as assuming that the system is initially relaxed. Therefore, an initially relaxed system described by a difference equation of the form in Equation (1.63) has the highly desirable linearity, time invariance, and causality properties. In this case, time invariance can be easily inferred if we consider that, for an initially relaxed system, the history of the system up to the application of the excitation is the same irrespective of the time sample position at which the excitation is applied. This happens because the outputs are all zero up to, but not including, the time of the application of the excitation. Therefore, if time is measured having as a reference the time sample $n = n_0$, at which the input is applied, then the output will not depend on the reference n_0 , because the history of the system prior to n_0 is the same irrespective of n_0 . This is equivalent to saying that if the input is shifted by k samples, then the output is just shifted by k samples, the rest remaining unchanged, thus characterizing a time-invariant system.

1.4.1 Recursive $\times$ nonrecursive systems

Equation (1.63) can be rewritten, without loss of generality, considering that $a_0 = 1$, yielding

$$y(n) = - \sum_{i=1}^N a_i y(n-i) + \sum_{l=0}^M b_l x(n-l). \quad (1.85)$$

This equation can be interpreted as the output signal $y(n)$ being dependent both on samples of the input, $x(n), x(n-1), \dots, x(n-M)$, and on previous samples of the output $y(n-1), y(n-2), \dots, y(n-N)$. Then, in this general case, we say that the system is recursive, since, in order to compute the output, we need past samples of the output itself. When $a_1 = a_2 = \dots = a_N = 0$, then the output at sample n depends only on values of the input signal. In such a case, the system is called nonrecursive, being distinctively characterized by a difference equation of the form

$$y(n) = \sum_{l=0}^M b_l x(n-l). \quad (1.86)$$

If we compare Equation (1.86) with the expression for the convolution sum given in Equation (1.50), we see that Equation (1.86) corresponds to a discrete system with impulse response $h(l) = b_l$. Since b_l is defined only for l between 0 and M , we can say that $h(l)$ is nonzero only for $0 \leq l \leq M$. This implies that the system in Equation (1.86) has a finite-duration impulse response. Such discrete-time systems are often referred to as finite-duration impulse-response (FIR) filters.

In contrast, when $y(n)$ depends on its past values, as in Equation (1.85), we have that the impulse response of the discrete system, in general, might not be zero when $n \rightarrow \infty$. Therefore, recursive digital systems are often referred to as infinite-duration impulse-response (IIR) filters.

Example 1.7. Find the impulse response of the system

$$y(n) - \frac{1}{\alpha} y(n-1) = x(n) \quad (1.87)$$

supposing that it is initially relaxed.

Solution

Since the system is initially relaxed, then $y(n) = 0$, for $n \leq -1$. Hence, for $n = 0$, we have that

$$y(0) = \frac{1}{\alpha} y(-1) + \delta(0) = \delta(0) = 1. \quad (1.88)$$

For $n > 0$, we have that

$$y(n) = \frac{1}{\alpha} y(n-1) \quad (1.89)$$

and, therefore, $y(n)$ can be expressed as

$$y(n) = \left(\frac{1}{\alpha}\right)^n u(n). \quad (1.90)$$

Note that $y(n) \neq 0$ for all $n \geq 0$; that is, the impulse response has infinite length. $\triangle$

One should note that, in general, recursive systems have IIRs, although there are some cases when recursive systems have FIRs. Illustrations of such cases can be found in Example 1.11, Exercise 1.16, and Section 12.5.

1.5 Solving difference equations

Consider the following homogeneous difference equation:

$$\sum_{i=0}^N a_i y(n-i) = 0. \quad (1.91)$$

We start by deriving an important property of it. Let $y_1(n)$ and $y_2(n)$ be solutions to Equation (1.91). Then

$$\sum_{i=0}^N a_i y_1(n-i) = 0 \quad (1.92)$$

$$\sum_{i=0}^N a_i y_2(n-i) = 0. \quad (1.93)$$

Adding Equation (1.92) multiplied by c_1 to Equation (1.93) multiplied by c_2 we have that

$$\begin{aligned} c_1 \sum_{i=0}^N a_i y_1(n-i) + c_2 \sum_{i=0}^N a_i y_2(n-i) &= 0 \\ \Rightarrow \sum_{i=0}^N a_i c_1 y_1(n-i) + \sum_{i=0}^N a_i c_2 y_2(n-i) &= 0 \\ \Rightarrow \sum_{i=0}^N a_i (c_1 y_1(n-i) + c_2 y_2(n-i)) &= 0. \end{aligned} \quad (1.94)$$

Equation (1.94) means that $(c_1 y_1(n) + c_2 y_2(n))$ is also a solution to Equation (1.91). This implies that, if $y_i(n)$, for $i = 0, 1, \dots, (M-1)$, are solutions of an homogeneous difference

equation, then

$$y_h(n) = \sum_{i=0}^{M-1} c_i y_i(n) \quad (1.95)$$

is also a solution.

As we have seen in Example 1.4, a difference equation may have solutions of the form

$$y(n) = K\rho^n. \quad (1.96)$$

Supposing that $y(n)$ from Equation (1.96) is also a solution to the difference equation (1.91), we have that

$$\sum_{i=0}^N a_i K\rho^{n-i} = 0. \quad (1.97)$$

If we disregard the trivial solution $\rho = 0$ and divide the left-hand side of Equation (1.97) by $K\rho^n$, we get

$$\sum_{i=0}^N a_i \rho^{-i} = 0, \quad (1.98)$$

which has the same solutions as the following polynomial equation:

$$\sum_{i=0}^N a_i \rho^{N-i} = 0. \quad (1.99)$$

As a result, one can conclude that if $\rho_0, \rho_1, \dots, \rho_{M-1}$, for $M \leq N$, are distinct zeros of the so-called characteristic polynomial in Equation (1.99), then there are M solutions for the homogeneous difference equation given by

$$y(n) = c_k \rho_k^n, \quad k = 0, 1, \dots, (M-1). \quad (1.100)$$

In fact, from Equation (1.95), we have that any linear combination of these solutions is also a solution for the homogeneous difference equation. Then, a homogeneous solution can be written as

$$y_h(n) = \sum_{k=0}^{M-1} c_k \rho_k^n, \quad (1.101)$$

where c_k , for $k = 0, 1, \dots, (M-1)$, are arbitrary constants.

Example 1.8. Find the general solution for the Fibonacci equation

$$y(n) = y(n-1) + y(n-2) \quad (1.102)$$

with $y(0) = 0$ and $y(1) = 1$.

Solution

The characteristic polynomial to the Fibonacci equation is

$$\rho^2 - \rho - 1 = 0 \quad (1.103)$$

whose roots are $\rho = (1 \pm \sqrt{5})/2$, leading to the general solution

$$y(n) = c_1 \left(\frac{1 + \sqrt{5}}{2} \right)^n + c_2 \left(\frac{1 - \sqrt{5}}{2} \right)^n. \quad (1.104)$$

Applying the auxiliary conditions $y(0) = 0$ and $y(1) = 1$ to Equation (1.104), we have that

$$\begin{cases} y(0) = c_1 + c_2 = 0 \\ y(1) = \left(\frac{1 + \sqrt{5}}{2} \right) c_1 + \left(\frac{1 - \sqrt{5}}{2} \right) c_2 = 1. \end{cases} \quad (1.105)$$

Thus, $c_1 = 1/\sqrt{5}$ and $c_2 = -1/\sqrt{5}$, and the solution to the Fibonacci equation becomes

$$y(n) = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]. \quad (1.106)$$

△

If the characteristic polynomial in Equation (1.99) has a pair of complex conjugate roots ρ and ρ^* of the form $a \pm jb = r e^{\pm j\phi}$, the associated homogeneous solution is given by

$$\begin{aligned} y_h(n) &= \hat{c}_1 (r e^{j\phi})^n + \hat{c}_2 (r e^{-j\phi})^n \\ &= r^n (\hat{c}_1 e^{j\phi n} + \hat{c}_2 e^{-j\phi n}) \\ &= r^n [(\hat{c}_1 + \hat{c}_2) \cos(\phi n) + j(\hat{c}_1 - \hat{c}_2) \sin(\phi n)] \\ &= c_1 r^n \cos(\phi n) + c_2 r^n \sin(\phi n). \end{aligned} \quad (1.107)$$

If the characteristic polynomial in Equation (1.99) has multiple roots, solutions distinct from Equation (1.100) are required. For example, if ρ is a double root, then there also exists a solution of the form

$$y_h(n) = cn\rho^n, \quad (1.108)$$

where c is an arbitrary constant. In general, if ρ is a root of multiplicity m , then the associated solution is of the form (Gabel & Roberts, 1980)

$$y_h(n) = \sum_{l=0}^{m-1} d_l n^l \rho^n, \quad (1.109)$$

where d_l , for $l = 0, 1, \dots, (m - 1)$, are arbitrary constants.

Table 1.1. Typical homogeneous solutions.

Root type [multiplicity]	Homogeneous solution $y_h(n)$
Real ρ_k [1]	$c_k \rho_k^n$
Real ρ_k [m_k]	$\sum_{l=0}^{m_k-1} d_l n^l \rho_k^n$
Complex conjugates $\rho_k, \rho_k^* = r e^{\pm j\phi}$ [1]	$r^n [c_1 \cos(\phi n) + c_2 \sin(\phi n)]$
Complex conjugates $\rho_k, \rho_k^* = r e^{\pm j\phi}$ [m_k]	$\sum_{l=0}^{m_k-1} [d_{1,l} n^l r^n \cos(\phi n) + d_{2,l} n^l r^n \sin(\phi n)]$

From the above, we can conclude that the homogeneous solutions of difference equations for each root type of the characteristic polynomial follow the rules summarized in Table 1.1.

A widely used method to find a particular solution for a difference equation of the form

$$\sum_{i=0}^N a_i y_p(n-i) = \sum_{l=0}^M b_l x(n-l) \quad (1.110)$$

is the so-called method of undetermined coefficients. This method can be used when the input sequence is the solution of a difference equation with constant coefficients. In order to do so, we define a delay operator $D\{\cdot\}$ as follows:

$$D^{-i}\{y(n)\} = y(n-i). \quad (1.111)$$

Such a delay operator is linear, since

$$\begin{aligned} D^{-i}\{c_1 y_1(n) + c_2 y_2(n)\} &= c_1 y_1(n-i) + c_2 y_2(n-i) \\ &= c_1 D^{-i}\{y_1(n)\} + c_2 D^{-i}\{y_2(n)\}. \end{aligned} \quad (1.112)$$

Also, the cascade of delay operators satisfies the following:

$$D^{-i}\{D^{-j}\{y(n)\}\} = D^{-i}\{y(n-j)\} = y(n-i-j) = D^{-(i+j)}\{y(n)\}. \quad (1.113)$$

Using delay operators, Equation (1.110) can be rewritten as

$$\left(\sum_{i=0}^N a_i D^{-i} \right) \{y_p(n)\} = \left(\sum_{l=0}^M b_l D^{-l} \right) \{x(n)\}. \quad (1.114)$$

The key idea is to find a difference operator $Q(D)$ of the form

$$Q(D) = \sum_{k=0}^R d_k D^{-k} = \prod_{r=0}^R (1 - \alpha_r D^{-1}) \quad (1.115)$$

Table 1.2. Annihilator polynomials for different input signals.

Input $x(n)$	Polynomial $Q(D)$
s^n	$1 - sD^{-1}$
n^i	$(1 - D^{-1})^{i+1}$
$n^i s^n$	$(1 - sD^{-1})^{i+1}$
$\cos(\omega n)$ or $\sin(\omega n)$	$(1 - e^{j\omega} D^{-1})(1 - e^{-j\omega} D^{-1})$
$s^n \cos(\omega n)$ or $s^n \sin(\omega n)$	$(1 - s e^{j\omega} D^{-1})(1 - s e^{-j\omega} D^{-1})$
$n \cos(\omega n)$ or $n \sin(\omega n)$	$[(1 - e^{j\omega} D^{-1})(1 - e^{-j\omega} D^{-1})]^2$

such that it annihilates the excitation; that is:

$$Q(D)\{x(n)\} = 0. \quad (1.116)$$

Applying $Q(D)$ to Equation (1.114) we get

$$\begin{aligned}
 Q(D) \left\{ \left(\sum_{i=0}^N a_i D^{-i} \right) \{y_p(n)\} \right\} &= Q(D) \left\{ \left(\sum_{l=0}^M b_l D^{-l} \right) \{x(n)\} \right\} \\
 &= \left(\sum_{l=0}^M b_l D^{-l} \right) \{Q(D)\{x(n)\}\} \\
 &= 0.
 \end{aligned} \quad (1.117)$$

This allows the nonhomogeneous difference equation to be solved using the same procedures used to find the homogeneous solutions.

For example, for a sequence $x(n) = s^n$, we have that $x(n-1) = s^{n-1}$; then:

$$x(n) = sx(n-1) \Rightarrow [1 - sD^{-1}]\{x(n)\} = 0$$

and, therefore, the annihilator polynomial for $x(n) = s^n$ is $Q(D) = (1 - sD^{-1})$. The annihilator polynomials for some typical inputs are summarized in Table 1.2.

Using the concept of annihilator polynomials, we can determine the form of the particular solution for certain types of input signal, which may include some undetermined coefficients. Some useful cases are shown in Table 1.3.

It is important to notice that there are no annihilator polynomials for inputs containing $u(n-n_0)$ or $\delta(n-n_0)$. Therefore, if a difference equation has inputs such as these, the above techniques can only be used for either $n \geq n_0$ or $n < n_0$, as discussed in Example 1.9.

Table 1.3. Typical particular solutions for different input signals.

Input $x(n)$	Particular solution $y_p(n)$
$s^n, s \neq \rho_k$	αs^n
$s^n, s = \rho_k$ with multiplicity m_k	$\alpha n^{m_k} s^n$
$\cos(\omega n + \phi)$	$\alpha \cos(\omega n + \phi)$
$\left(\sum_{i=0}^I \beta_i n^i \right) s^n$	$\left(\sum_{i=0}^I \alpha_i n^i \right) s^n$

Example 1.9. Solve the difference equation

$$y(n) + a^2 y(n-2) = b^n \sin\left(\frac{\pi}{2}n\right) u(n) \quad (1.118)$$

assuming that $a \neq b$ and $y(n) = 0$, for $n < 0$.

Solution

Using operator notation, Equation (1.118) becomes

$$(1 + a^2 D^{-2})\{y(n)\} = b^n \sin\left(\frac{\pi}{2}n\right) u(n). \quad (1.119)$$

The homogeneous equation is

$$y_h(n) + a^2 y_h(n-2) = 0. \quad (1.120)$$

Then, the characteristic polynomial equation from which we derive the homogeneous solution is

$$\rho^2 + a^2 = 0. \quad (1.121)$$

Since its roots are $\rho = a e^{\pm j\pi/2}$, then two solutions for the homogeneous equation are $a^n \sin[(\pi/2)n]$ and $a^n \cos[(\pi/2)n]$, as given in Table 1.1. Then, the general homogeneous solution becomes

$$y_h(n) = a^n \left[c_1 \sin\left(\frac{\pi}{2}n\right) + c_2 \cos\left(\frac{\pi}{2}n\right) \right]. \quad (1.122)$$

If one applies the correct annihilation to the excitation signals, the original difference equation is transformed into a higher order homogeneous equation. The solutions of this higher order homogeneous equation include the homogeneous and particular solutions of the original difference equation. However, there is no annihilator polynomial for $b^n \sin[(\pi/2)n]u(n)$. Therefore, one can only compute the solution to the difference equation

for $n \geq 0$, when the term to be annihilated becomes just $b^n \sin[(\pi/2)n]$. Therefore, for $n \geq 0$, according to Table 1.2, for the given input signal the annihilator polynomial is given by

$$Q(D) = (1 - b e^{j\pi/2} D^{-1})(1 - b e^{-j\pi/2} D^{-1}) = 1 + b^2 D^{-2}. \quad (1.123)$$

Applying the annihilator polynomial on the difference equation, we obtain²

$$(1 + b^2 D^{-2})(1 + a^2 D^{-2})\{y(n)\} = 0. \quad (1.124)$$

The corresponding polynomial equation is

$$(\rho^2 + b^2)(\rho^2 + a^2) = 0. \quad (1.125)$$

It has four roots, two of the form $\rho = a e^{\pm j\pi/2}$ and two of the form $\rho = b e^{\pm j\pi/2}$. Since $a \neq b$, for $n \geq 0$ the complete solution is then given by

$$y(n) = b^n \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] + a^n \left[d_3 \sin\left(\frac{\pi}{2}n\right) + d_4 \cos\left(\frac{\pi}{2}n\right) \right]. \quad (1.126)$$

The constants d_i , for $i = 1, 2, 3, 4$, are computed such that $y(n)$ is a particular solution to the nonhomogeneous equation. However, we notice that the term involving a^n corresponds to the solution of the homogeneous equation. Therefore, we do not need to substitute it in the equation since it will be annihilated for every d_3 and d_4 . One can then compute d_1 and d_2 by substituting only the term involving b^n in the nonhomogeneous Equation (1.118), leading to the following algebraic development:

$$\begin{aligned} & b^n \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] \\ & + a^2 b^{n-2} \left\{ d_1 \sin\left[\frac{\pi}{2}(n-2)\right] + d_2 \cos\left[\frac{\pi}{2}(n-2)\right] \right\} = b^n \sin\left(\frac{\pi}{2}n\right) \\ \Rightarrow & \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] \\ & + a^2 b^{-2} \left[d_1 \sin\left(\frac{\pi}{2}n - \pi\right) + d_2 \cos\left(\frac{\pi}{2}n - \pi\right) \right] = \sin\left(\frac{\pi}{2}n\right) \\ \Rightarrow & \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] \\ & + a^2 b^{-2} \left[-d_1 \sin\left(\frac{\pi}{2}n\right) - d_2 \cos\left(\frac{\pi}{2}n\right) \right] = \sin\left(\frac{\pi}{2}n\right) \\ \Rightarrow & d_1(1 - a^2 b^{-2}) \sin\left(\frac{\pi}{2}n\right) + d_2(1 - a^2 b^{-2}) \cos\left(\frac{\pi}{2}n\right) = \sin\left(\frac{\pi}{2}n\right). \end{aligned} \quad (1.127)$$

Therefore, we conclude that

$$d_1 = \frac{1}{1 - a^2 b^{-2}}; \quad d_2 = 0 \quad (1.128)$$

² Since the expression of the input is valid only for $n \geq 0$, in this case, technically speaking, the annihilator polynomial should have a noncausal form with only nonnegative exponents for the delay operator, resulting in $Q(D) = D^2 + b^2$. In practice, however, the two expressions present the same roots and, therefore, they are equivalent and can be readily interchanged, as suggested here.

and the overall solution for $n \geq 0$ is

$$y(n) = \frac{b^n}{1 - a^2 b^{-2}} \sin\left(\frac{\pi}{2}n\right) + a^n \left[d_3 \sin\left(\frac{\pi}{2}n\right) + d_4 \cos\left(\frac{\pi}{2}n\right) \right]. \quad (1.129)$$

We now compute the constants d_3 and d_4 using the auxiliary conditions generated by the condition $y(n) = 0$, for $n < 0$. This implies that $y(-1) = 0$ and $y(-2) = 0$. However, one cannot use Equation (1.129) since it is valid only for $n \geq 0$. Thus, we need to run the difference equation from the auxiliary conditions $y(-2) = y(-1) = 0$ to compute $y(0)$ and $y(1)$:

$$\begin{cases} n = 0 : & y(0) + a^2 y(-2) = b^0 \sin\left(\frac{\pi}{2} \times 0\right) u(0) = 0 \\ n = 1 : & y(1) + a^2 y(-1) = b^1 \sin\left(\frac{\pi}{2}\right) u(1) = b \end{cases} \Rightarrow \begin{cases} y(0) = 0 \\ y(1) = b. \end{cases} \quad (1.130)$$

Using these auxiliary conditions in Equation (1.129) we get

$$\begin{cases} y(0) = \frac{1}{1 - a^2 b^{-2}} \sin\left(\frac{\pi}{2} \times 0\right) + \left[d_3 \sin\left(\frac{\pi}{2} \times 0\right) + d_4 \cos\left(\frac{\pi}{2} \times 0\right) \right] = 0 \\ y(1) = \frac{b}{1 - a^2 b^{-2}} \sin\left(\frac{\pi}{2}\right) + a \left[d_3 \sin\left(\frac{\pi}{2}\right) + d_4 \cos\left(\frac{\pi}{2}\right) \right] = b \end{cases} \quad (1.131)$$

and then

$$\begin{cases} d_4 = 0 \\ \frac{b}{1 - a^2 b^{-2}} + a d_3 = b \end{cases} \Rightarrow \begin{cases} d_3 = -\frac{ab^{-1}}{1 - a^2 b^{-2}} \\ d_4 = 0. \end{cases} \quad (1.132)$$

Substituting these values in Equation (1.129), the general solution becomes

$$\begin{cases} y(n) = 0, & n < 0 \\ y(n) = \frac{b^n - a^{n+1} b^{-1}}{1 - a^2 b^{-2}} \sin\left(\frac{\pi}{2}n\right), & n \geq 0, \end{cases} \quad (1.133)$$

which can be expressed in compact form as

$$y(n) = \frac{b^n - a^{n+1} b^{-1}}{1 - a^2 b^{-2}} \sin\left(\frac{\pi}{2}n\right) u(n). \quad (1.134)$$

An interesting case arises if the excitation is a pure sinusoid; that is, if $a = 1$, then the above solution can be written as

$$y(n) = \frac{b^n}{1 - b^{-2}} \sin\left(\frac{\pi}{2}n\right) u(n) - \frac{b^{-1}}{1 - b^{-2}} \sin\left(\frac{\pi}{2}n\right) u(n). \quad (1.135)$$

If $b > 1$, for large values of n , the first term of the right-hand side grows without bound (since b^n tends to infinity) and, therefore, the system is unstable (see Section 1.3.5). On the other hand, if $b < 1$, then b^n tends to zero as n grows and, therefore, the solution becomes the pure sinusoid

$$y(n) = -\frac{b^{-1}}{1 - b^{-2}} \sin\left(\frac{\pi}{2}n\right). \quad (1.136)$$

We refer to this as a steady-state solution of the difference equation (see Exercises 1.17 and 1.18). Such solutions are very important in practice, and in Chapter 2 other techniques to compute them will be studied. $\triangle$

Example 1.10. Determine the solution of the difference equation in Example 1.9 supposing that $a = b$ (observe that the annihilator polynomial has common zeros with the homogeneous equation).

Solution

For $a = b$, there are repeated roots in the difference equation, and as a result the complete solution has the following form for $n \geq 0$:

$$y(n) = na^n \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] + a^n \left[d_3 \sin\left(\frac{\pi}{2}n\right) + d_4 \cos\left(\frac{\pi}{2}n\right) \right]. \quad (1.137)$$

As in the case for $a \neq b$, we notice that the right-hand side of the summation is the homogeneous solution, and thus it will be annihilated for all d_3 and d_4 . For finding d_1 and d_2 one should substitute the left-hand side of the summation in the original equation (1.118), for $n \geq 0$. This yields

$$\begin{aligned} & na^n \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] \\ & + a^2(n-2)a^{n-2} \left\{ d_1 \sin\left[\frac{\pi}{2}(n-2)\right] + d_2 \cos\left[\frac{\pi}{2}(n-2)\right] \right\} = a^n \sin\left(\frac{\pi}{2}n\right) \\ \Rightarrow & n \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] \\ & + (n-2) \left[d_1 \sin\left(\frac{\pi}{2}n - \pi\right) + d_2 \cos\left(\frac{\pi}{2}n - \pi\right) \right] = \sin\left(\frac{\pi}{2}n\right) \\ \Rightarrow & n \left[d_1 \sin\left(\frac{\pi}{2}n\right) + d_2 \cos\left(\frac{\pi}{2}n\right) \right] \\ & + (n-2) \left[-d_1 \sin\left(\frac{\pi}{2}n\right) - d_2 \cos\left(\frac{\pi}{2}n\right) \right] = \sin\left(\frac{\pi}{2}n\right) \\ \Rightarrow & [nd_1 - (n-2)d_1] \sin\left(\frac{\pi}{2}n\right) + [nd_2 - (n-2)d_2] \cos\left(\frac{\pi}{2}n\right) = \sin\left(\frac{\pi}{2}n\right) \\ \Rightarrow & 2d_1 \sin\left(\frac{\pi}{2}n\right) + 2d_2 \cos\left(\frac{\pi}{2}n\right) = \sin\left(\frac{\pi}{2}n\right). \end{aligned} \quad (1.138)$$

Therefore, we conclude that

$$d_1 = \frac{1}{2}; \quad d_2 = 0 \quad (1.139)$$

and the overall solution for $n \geq 0$ is

$$y(n) = \frac{na^n}{2} \sin\left(\frac{\pi}{2}n\right) + a^n \left[d_3 \sin\left(\frac{\pi}{2}n\right) + d_4 \cos\left(\frac{\pi}{2}n\right) \right]. \quad (1.140)$$

As in the case for $a \neq b$, in order to compute the constants d_3 and d_4 one must use auxiliary conditions for $n \geq 0$, since Equation (1.137) is only valid for $n \geq 0$. Since $y(n) = 0$, for $n < 0$, we need to run the difference equation from the auxiliary conditions $y(-2) = y(-1) = 0$ to compute $y(0)$ and $y(1)$:

$$\begin{cases} n = 0 : & y(0) + a^2 y(-2) = a^0 \sin\left(\frac{\pi}{2} \times 0\right) u(0) = 0 \\ n = 1 : & y(1) + a^2 y(-1) = a^1 \sin\left(\frac{\pi}{2}\right) u(1) = a \end{cases} \Rightarrow \begin{cases} y(0) = 0 \\ y(1) = a. \end{cases} \quad (1.141)$$

Using these auxiliary conditions in Equation (1.140), we get

$$\begin{cases} y(0) = d_4 = 0 \\ y(1) = a \left[\frac{1}{2} \sin\left(\frac{\pi}{2}\right) \right] + a \left[d_3 \sin\left(\frac{\pi}{2}\right) + d_4 \cos\left(\frac{\pi}{2}\right) \right] = a \end{cases} \quad (1.142)$$

and then

$$\frac{a}{2} + ad_3 = a \Rightarrow d_3 = \frac{1}{2}; \quad d_4 = 0 \quad (1.143)$$

and since $y(n) = 0$, for $n < 0$, the solution is

$$y(n) = \frac{n+1}{2} a^n \sin\left(\frac{\pi}{2}n\right) u(n). \quad (1.144)$$

△

1.5.1 Computing impulse responses

In order to find the impulse response of a system, we can start by solving the following difference equation:

$$\sum_{i=0}^N a_i y(n-i) = \delta(n). \quad (1.145)$$

As has been pointed out in the discussion preceding Example 1.6, for a linear system to be causal it must be initially relaxed; that is, the auxiliary conditions prior to the input must be zero. For causal systems, since the input $\delta(n)$ is applied at $n = 0$, we must have

$$y(-1) = y(-2) = \cdots = y(-N) = 0. \quad (1.146)$$

For $n > 0$, Equation (1.145) becomes homogeneous; that is

$$\sum_{i=0}^N a_i y(n-i) = 0. \quad (1.147)$$

This can be solved using the techniques presented earlier in Section 1.5. In order to do so, we need N auxiliary conditions. However, since Equation (1.147) is valid only for $n > 0$, we cannot use the auxiliary conditions from Equation (1.146), but need N auxiliary conditions for $n > 0$ instead. For example, these conditions can be $y(1), y(2), \dots, y(N)$, which can be found, starting from the auxiliary conditions in Equation (1.146), by running the difference Equation (1.145) from $n = 0$ to $n = N$, leading to

$$\begin{cases} n = 0 : y(0) = \frac{\delta(0)}{a_0} - \frac{1}{a_0} \sum_{i=1}^N a_i y(-i) = \frac{1}{a_0} \\ n = 1 : y(1) = \frac{\delta(1)}{a_0} - \frac{1}{a_0} \sum_{i=1}^N a_i y(1-i) = -\frac{a_1}{a_0^2} \\ \vdots \\ n = N : y(N) = \frac{\delta(N)}{a_0} - \frac{1}{a_0} \sum_{i=1}^N a_i y(N-i) = -\frac{1}{a_0} \sum_{i=1}^N a_i y(N-i). \end{cases} \quad (1.148)$$

Example 1.11. Compute the impulse response of the system governed by the following difference equation:

$$y(n) - \frac{1}{2}y(n-1) + \frac{1}{4}y(n-2) = x(n). \quad (1.149)$$

Solution

For $n > 0$ the impulse response satisfies the homogeneous equation. The corresponding polynomial equation is

$$\rho^2 - \frac{1}{2}\rho + \frac{1}{4} = 0 \quad (1.150)$$

whose roots are $\rho = \frac{1}{2} e^{\pm j\pi/3}$. Therefore, for $n > 0$, the solution is

$$y(n) = c_1 2^{-n} \cos\left(\frac{\pi}{3}n\right) + c_2 2^{-n} \sin\left(\frac{\pi}{3}n\right). \quad (1.151)$$

Considering the system to be causal, we have that $y(n) = 0$, for $n < 0$. Therefore, we need to compute the auxiliary conditions for $n > 0$ as follows:

$$\begin{cases} n = 0 : y(0) = \delta(0) + \frac{1}{2}y(-1) - \frac{1}{4}y(-2) = 1 \\ n = 1 : y(1) = \delta(1) + \frac{1}{2}y(0) - \frac{1}{4}y(-1) = \frac{1}{2} \\ n = 2 : y(2) = \delta(2) + \frac{1}{2}y(1) - \frac{1}{4}y(0) = 0. \end{cases} \quad (1.152)$$

Applying the above conditions to the solution in Equation (1.151) we have

$$\begin{cases} y(1) = c_1 2^{-1} \cos\left(\frac{\pi}{3}\right) + c_2 2^{-1} \sin\left(\frac{\pi}{3}\right) = \frac{1}{2} \\ y(2) = c_1 2^{-2} \cos\left(\frac{2\pi}{3}\right) + c_2 2^{-2} \sin\left(\frac{2\pi}{3}\right) = 0. \end{cases} \quad (1.153)$$

Hence:

$$\begin{cases} \frac{1}{4}c_1 + \frac{\sqrt{3}}{4}c_2 = \frac{1}{2} \\ -\frac{1}{8}c_1 + \frac{\sqrt{3}}{8}c_2 = 0 \end{cases} \Rightarrow \begin{cases} c_1 = 1 \\ c_2 = \frac{\sqrt{3}}{3} \end{cases} \quad (1.154)$$

and the impulse response becomes

$$y(n) = \begin{cases} 0, & n < 0 \\ 1, & n = 0 \\ \frac{1}{2}, & n = 1 \\ 0, & n = 2 \\ 2^{-n} \left\{ \cos[(\pi/3)n] + (\sqrt{3}/3) \sin[(\pi/3)n] \right\}, & n \geq 2 \end{cases} \quad (1.155)$$

which, by inspection, can be expressed in a compact form as

$$y(n) = 2^{-n} \left[\cos\left(\frac{\pi}{3}n\right) + \frac{\sqrt{3}}{3} \sin\left(\frac{\pi}{3}n\right) \right] u(n). \quad (1.156)$$

△

1.6 Sampling of continuous-time signals

In many cases, a discrete-time signal $x(n)$ consists of samples of a continuous-time signal $x_a(t)$; that is:

$$x(n) = x_a(nT). \quad (1.157)$$

If we want to process the continuous-time signal $x_a(t)$ using a discrete-time system, then we first need to convert it using Equation (1.157), process the discrete-time input digitally, and then convert the discrete-time output back to the continuous-time domain. Therefore, in order for this operation to be effective, it is essential that we have capability of restoring a continuous-time signal from its samples. In this section, we derive conditions under which a continuous-time signal can be recovered from its samples. We also devise ways of performing this recovery. To do so, we first introduce some basic concepts of analog signal processing that can be found in most standard books on linear systems (Gabel & Roberts, 1980; Oppenheim *et al.*, 1983). Following that, we derive the sampling theorem, which gives the basis for digital processing of continuous-time signals.

1.6.1 Basic principles

The Fourier transform of a continuous-time signal $f(t)$ is given by

$$F(j\Omega) = \int_{-\infty}^{\infty} f(t) e^{-j\Omega t} dt, \quad (1.158)$$

where Ω is referred to as the frequency and is measured in radians per second (rad/s). The corresponding inverse relationship is expressed as

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\Omega) e^{j\Omega t} d\Omega. \quad (1.159)$$

An important property associated with the Fourier transform is that the Fourier transform of the product of two functions equals the convolution of their Fourier transforms. That is, if $x(t) = a(t)b(t)$, then

$$X(j\Omega) = \frac{1}{2\pi} A(j\Omega) * B(j\Omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} A(j\Omega - j\Omega') B(j\Omega') d\Omega', \quad (1.160)$$

where $X(j\Omega)$, $A(j\Omega)$, and $B(j\Omega)$ are the Fourier transforms of $x(t)$, $a(t)$, and $b(t)$ respectively.

In addition, if a signal $x(t)$ is periodic with period T , then we can express it by its Fourier series defined by

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j(2\pi/T)kt}, \quad (1.161)$$

where the a_k are called the series coefficients, which are determined as

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk(2\pi/T)t} dt. \quad (1.162)$$

1.6.2 Sampling theorem

Given a discrete-time signal $x(n)$ derived from a continuous-time signal $x_a(t)$ using Equation (1.157), we define a continuous-time signal $x_i(t)$ consisting of a train of impulses at $t = nT$, each of area equal to $x(n) = x_a(nT)$. Examples of signals $x_a(t)$, $x(n)$, and $x_i(t)$ are depicted in Figure 1.5, where the direct relationships between these three signals can be seen.

The signal $x_i(t)$ can be expressed as

$$x_i(t) = \sum_{n=-\infty}^{\infty} x(n) \delta(t - nT). \quad (1.163)$$

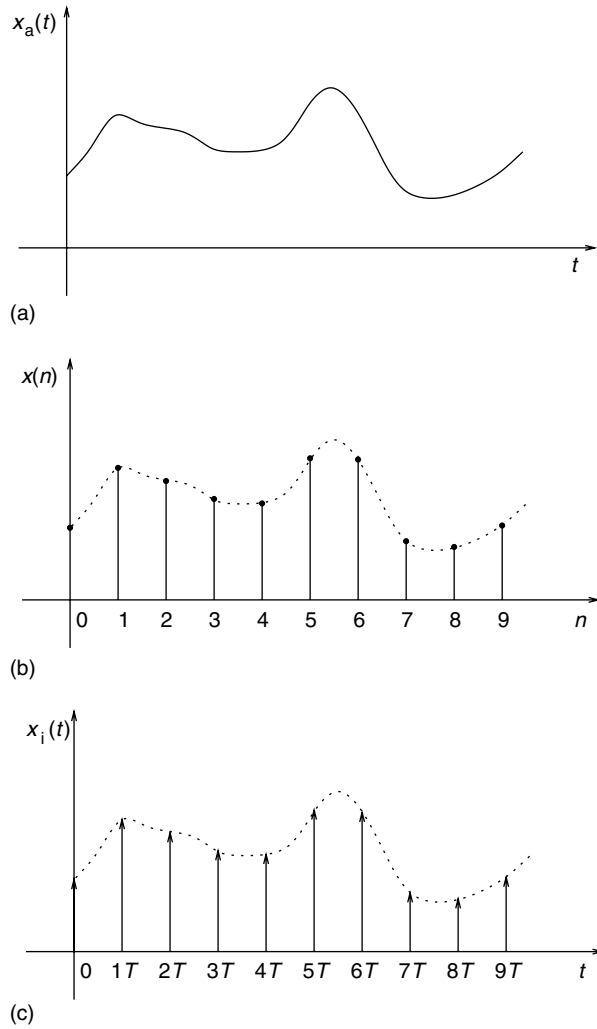


Fig. 1.5.

(a) Continuous-time signal $x_a(t)$; (b) discrete-time signal $x(n)$; (c) auxiliary continuous-time signal $x_i(t)$.

Since, from Equation (1.157), $x(n) = x_a(nT)$, then Equation (1.163) becomes

$$x_i(t) = \sum_{n=-\infty}^{\infty} x_a(nT) \delta(t - nT) = x_a(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = x_a(t) p(t), \quad (1.164)$$

indicating that $x_i(t)$ can also be obtained by multiplying the continuous-time signal $x_a(t)$ by a train of impulses $p(t)$ defined as

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT). \quad (1.165)$$

In the equations above, we have defined a continuous-time signal $x_i(t)$ that can be obtained from the discrete-time signal $x(n)$ in a straightforward manner. In what follows, we relate the Fourier transforms of $x_a(t)$ and $x_i(t)$, and study the conditions under which $x_a(t)$ can be obtained from $x_i(t)$.

From equations (1.160) and (1.164), the Fourier transform of $x_i(t)$ is such that

$$X_i(j\Omega) = \frac{1}{2\pi} X_a(j\Omega) * P(j\Omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_a(j\Omega - j\Omega') P(j\Omega') d\Omega'. \quad (1.166)$$

Therefore, in order to arrive at an expression for the Fourier transform of $x_i(t)$, we must first determine the Fourier transform of $p(t)$, $P(j\Omega)$. From Equation (1.165), we see that $p(t)$ is a periodic function with period T and that we can decompose it in a Fourier series, as described in Equations (1.161) and (1.162). Since, from Equation (1.165), $p(t)$ in the interval $[-T/2, T/2]$ is equal to just an impulse $\delta(t)$, the coefficients a_k in the Fourier series of $p(t)$ are given by

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk(2\pi/T)t} dt = \frac{1}{T} \quad (1.167)$$

and the Fourier series for $p(t)$ becomes

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j(2\pi/T)kt}. \quad (1.168)$$

As the Fourier transform of $f(t) = e^{j\Omega_0 t}$ is equal to $F(j\Omega) = 2\pi\delta(\Omega - \Omega_0)$, then, from Equation (1.168), the Fourier transform of $p(t)$ becomes

$$P(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\Omega - \frac{2\pi}{T}k\right). \quad (1.169)$$

Substituting this expression for $P(j\Omega)$ in Equation (1.166), we have that

$$\begin{aligned} X_i(j\Omega) &= \frac{1}{2\pi} X_a(j\Omega) * P(j\Omega) \\ &= \frac{1}{T} X_a(j\Omega) * \sum_{k=-\infty}^{\infty} \delta\left(\Omega - \frac{2\pi}{T}k\right) \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(j\Omega - j\frac{2\pi}{T}k\right), \end{aligned} \quad (1.170)$$

where, in the last step, we used the fact that the convolution of a function $F(j\Omega)$ with a shifted impulse $\delta(\Omega - \Omega_0)$ is the shifted function $F(j\Omega - j\Omega_0)$. Equation (1.170) shows that the spectrum of $x_i(t)$ is composed of infinite shifted copies of the spectrum of $x_a(t)$, with the shifts in frequency being multiples of the sampling frequency, $\Omega_s = 2\pi/T$. Figure 1.6 shows examples of signals $x_a(t)$, $p(t)$, and $x_i(t)$, and their respective Fourier transforms.

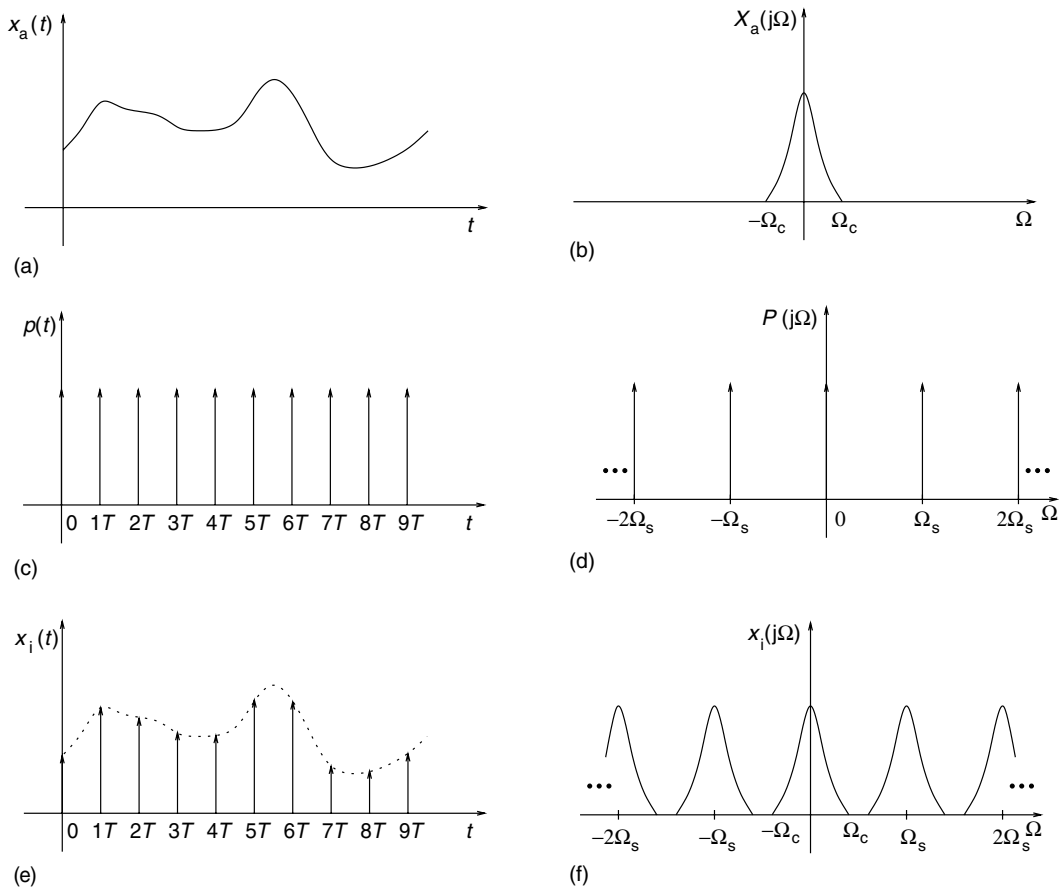


Fig. 1.6. (a) Continuous-time signal $x_a(t)$; (b) spectrum of $x_a(t)$; (c) train of impulses $p(t)$; (d) spectrum of $p(t)$; (e) auxiliary continuous-time signal $x_i(t)$; (f) spectrum of $x_i(t)$.

From Equation (1.170) and Figure 1.6f, we see that, in order to avoid the repeated copies of the spectrum of $x_a(t)$ interfering with one another, the signal should be band-limited. In addition, its bandwidth Ω_c should be such that the upper edge of the spectrum centered at zero is smaller than the lower edge of the spectrum centered at Ω_s .

Referring to the general complex case depicted in Figure 1.7, we must have $\Omega_s + \Omega_2 > \Omega_1$, or equivalently $\Omega_s > \Omega_1 - \Omega_2$.

In the case of real signals, since the spectrum is symmetric around zero, the one-sided bandwidth of the continuous-time signal Ω_c is such that $\Omega_c = \Omega_1 = -\Omega_2$, and then we must have $\Omega_s > \Omega_c - (-\Omega_c)$, implying that

$$\Omega_s > 2\Omega_c; \quad (1.171)$$

that is, the sampling frequency must be larger than double the one-sided bandwidth of the continuous-time signal. The frequency $\Omega = 2\Omega_c$ is called the Nyquist frequency of the real continuous-time signal $x_a(t)$.

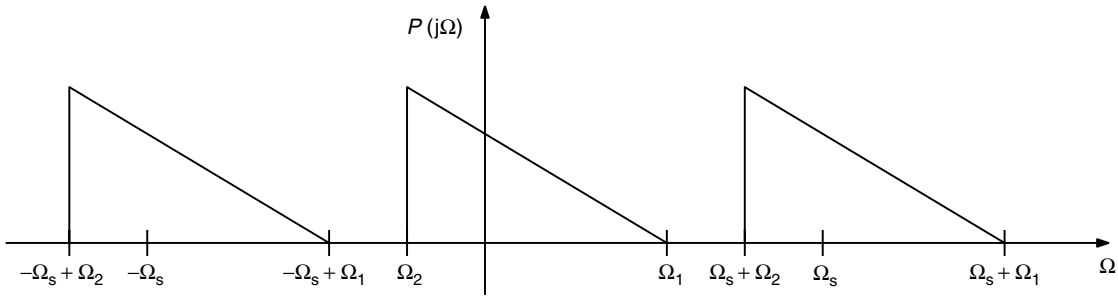


Fig. 1.7. Example of spectrum of a sampled complex signal.

In addition, if the condition in Equation (1.171) is satisfied, the original continuous signal $x_a(t)$ can be recovered by isolating the part of the spectrum of $x_i(t)$ that corresponds to the spectrum of $x_a(t)$.³ This can be achieved by filtering the signal $x_i(t)$ with an ideal lowpass filter having bandwidth $\Omega_s/2$.

On the other hand, if the condition in Equation (1.171) is not satisfied, the repetitions of the spectrum interfere with one another, and the continuous-time signal cannot be recovered from its samples. This superposition of the repetitions of the spectrum of $x_a(t)$ in $x_i(t)$, when the sampling frequency is smaller than $2\Omega_c$, is commonly referred to as aliasing. Figure 1.8b–d shows the spectra of $x_i(t)$ for Ω_s equal to, smaller than, and larger than $2\Omega_c$, respectively. The aliasing phenomenon is clearly identified in Figure 1.8c.

We are now ready to enunciate a very important result.

Theorem 1.1 (Sampling Theorem). *If a continuous-time signal $x_a(t)$ is band-limited – that is, its Fourier transform is such that $X_a(j\Omega) = 0$, for $|\Omega| > |\Omega_c|$ – then $x_a(t)$ can be completely recovered from the discrete-time signal $x(n) = x_a(nT)$ if the sampling frequency Ω_s satisfies $\Omega_s > 2\Omega_c$.*

◇

Example 1.12. Consider the discrete-time sequence

$$x(n) = \sin\left(\frac{6\pi}{4}n\right). \quad (1.172)$$

Assuming that the sampling frequency is $f_s = 40$ kHz, find two continuous-time signals that could have generated this sequence.

Solution

Supposing that the continuous-time signal is of the form

$$x_a(t) = \sin(\Omega_c t) = \sin(2\pi f_c t). \quad (1.173)$$

³ In fact, any of the spectrum repetitions have the full information about $x_a(t)$. However, if we isolate a repetition of the spectrum not centered at $\Omega = 0$, we get a modulated version of $x_a(t)$, which should be demodulated. Since its demodulation is equivalent to shifting the spectrum back to the origin, it is usually better to take the repetition of the spectrum centered at the origin in the first place.

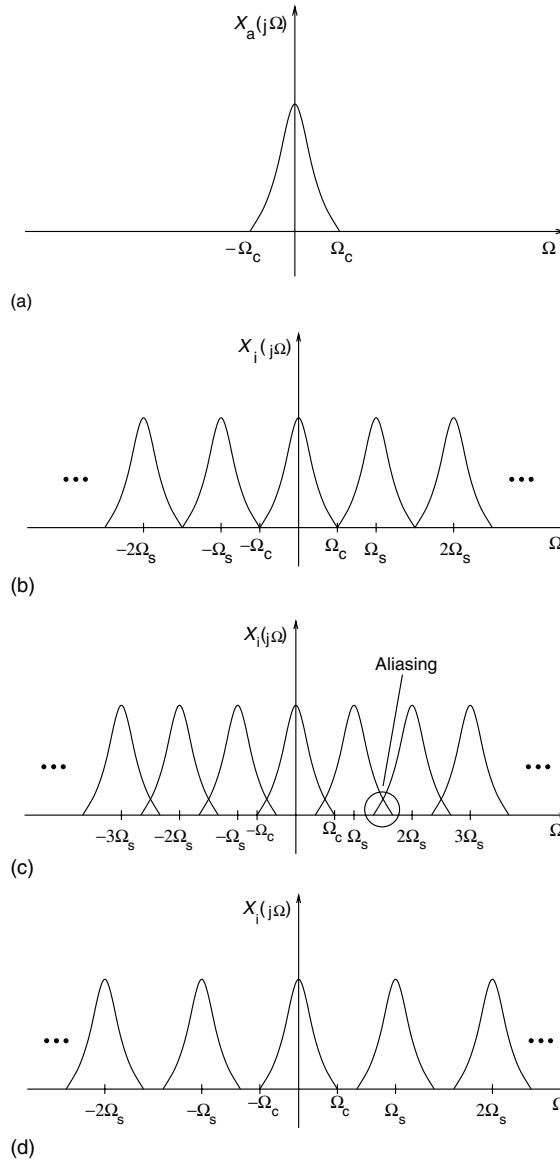


Fig. 1.8.

(a) Spectrum of the continuous-time signal. Spectra of $x_i(t)$ for: (b) $\Omega_s = 2\Omega_c$; (c) $\Omega_s < 2\Omega_c$; (d) $\Omega_s > 2\Omega_c$.

We have that when sampled with sampling frequency $f_s = 1/T_s$ it generates the following discrete signal:

$$\begin{aligned}
 x(n) &= x_a(nT_s) \\
 &= \sin(2\pi f_c nT_s) \\
 &= \sin\left(2\pi \frac{f_c}{f_s} n\right)
 \end{aligned}$$

$$\begin{aligned}
&= \sin\left(2\pi \frac{f_c}{f_s} n + 2k\pi n\right) \\
&= \sin\left[2\pi \left(\frac{f_c}{f_s} + k\right) n\right]
\end{aligned} \tag{1.174}$$

for any integer k . Therefore, in order for a sinusoid following Equation (1.173), when sampled, to yield the discrete signal in Equation (1.172), we must have that

$$2\pi \left(\frac{f_c}{f_s} + k\right) = \frac{6\pi}{4} \Rightarrow f_c = \left(\frac{3}{4} - k\right) f_s. \tag{1.175}$$

For example:

$$k = 0 \Rightarrow f_c = \frac{3}{4} f_s = 30 \text{ kHz} \Rightarrow x_1(t) = \sin(60\,000\pi t) \tag{1.176}$$

$$k = -1 \Rightarrow f_c = \frac{7}{4} f_s = 70 \text{ kHz} \Rightarrow x_2(t) = \sin(140\,000\pi t) \tag{1.177}$$

We can verify that by computing the $x_i(t)$ signals for the two above signals according to Equation (1.164):

$$\begin{aligned}
x_{1i}(t) &= \sum_{n=-\infty}^{\infty} x_1(t) \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} \sin(60\,000\pi t) \delta\left(t - \frac{n}{40\,000}\right) \\
&= \sum_{n=-\infty}^{\infty} \sin\left(60\,000\pi \frac{n}{40\,000}\right) \delta\left(t - \frac{n}{40\,000}\right) \\
&= \sum_{n=-\infty}^{\infty} \sin\left(\frac{3\pi}{2} n\right) \delta\left(t - \frac{n}{40\,000}\right)
\end{aligned} \tag{1.178}$$

$$\begin{aligned}
x_{2i}(t) &= \sum_{n=-\infty}^{\infty} x_2(t) \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} \sin(140\,000\pi t) \delta\left(t - \frac{n}{40\,000}\right) \\
&= \sum_{n=-\infty}^{\infty} \sin\left(140\,000\pi \frac{n}{40\,000}\right) \delta\left(t - \frac{n}{40\,000}\right) \\
&= \sum_{n=-\infty}^{\infty} \sin\left(\frac{7\pi}{2} n\right) \delta\left(t - \frac{n}{40\,000}\right).
\end{aligned} \tag{1.179}$$

Since

$$\sin\left(\frac{7\pi}{2} n\right) = \sin\left[\left(\frac{3\pi}{2} + 2\pi\right) n\right] = \sin\left(\frac{3\pi}{2} n\right), \tag{1.180}$$

then we have that the signals $x_{1i}(t)$ and $x_{2i}(t)$ are identical. $\triangle$

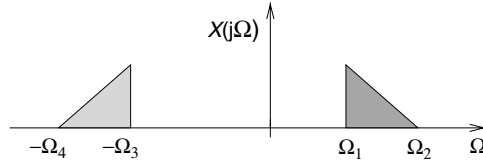


Fig. 1.9.

Signal spectrum of Example 1.13.

Example 1.13. In Figure 1.9, assuming that $\Omega_2 - \Omega_1 < \Omega_1$ and $\Omega_4 - \Omega_3 < \Omega_3$:

- Using a single sampler, what would be the minimum sampling frequency such that no information is lost?
- Using an ideal filter and two samplers, what would be the minimum sampling frequencies such that no information is lost? Depict the configuration used in this case.

Solution

- By examining Figure 1.9 we see that a sampling rate $\Omega_s > \Omega_2 + \Omega_4$ would avoid aliasing. However, since it is given that $\Omega_2 - \Omega_1 < \Omega_1$ and $\Omega_4 - \Omega_3 < \Omega_3$, then in the empty spectrum between $-\Omega_3$ and Ω_1 we can accommodate one copy of the spectrum in the interval $[\Omega_1, \Omega_2]$ and one copy of the spectrum in the interval $[-\Omega_4, -\Omega_3]$. According to Equation (1.170), when a signal is sampled its spectrum is repeated at multiples of Ω_s . Therefore, we can choose Ω_s so that the spectrum of the sampled signal would be as in the lower part of Figure 1.10, where, to avoid spectrum superposition, we must have

$$\begin{cases} \Omega_1 - \Omega_s > -\Omega_3 \\ -\Omega_4 + \Omega_s > \Omega_2 - \Omega_s \end{cases} \Rightarrow \frac{\Omega_2 + \Omega_4}{2} < \Omega_s < \Omega_1 + \Omega_3. \quad (1.181)$$

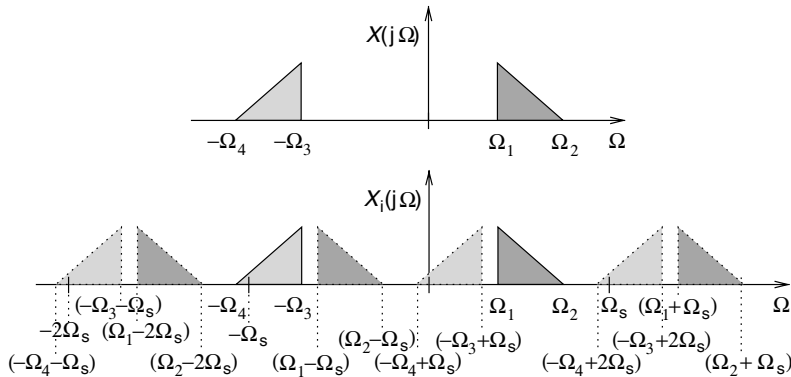


Fig. 1.10.

Spectrum of the sampled signal in Example 1.13(a).

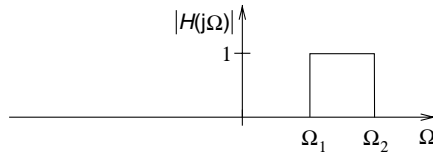


Fig. 1.11. Ideal bandpass filter.

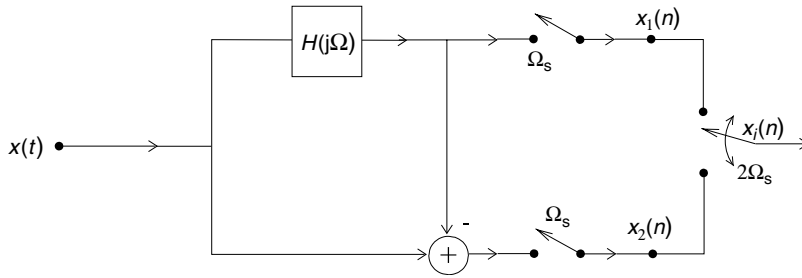


Fig. 1.12. Sampling solution in Example 1.13(b) using an ideal bandpass filter.

Therefore, the minimum sampling frequency would be $\Omega_s = (\Omega_2 + \Omega_4)/2$, provided that $\Omega_s < \Omega_1 + \Omega_3$.

- (b) If we have an ideal filter, as depicted in Figure 1.11, then we can isolate both parts of the spectrum and then sample them at a much lower rate. For example, we can sample the output of the filter in Figure 1.11 with a frequency $\Omega_{s1} > \Omega_2 - \Omega_1$. If we use the scheme in Figure 1.12, then we take the filter output and subtract it from the input signal. The result will have just the left side of the spectrum in Figure 1.9, which can be sampled with a frequency $\Omega_{s2} > \Omega_4 - \Omega_3$.

If we use a single sampling frequency, then its value should satisfy

$$\Omega_s > \max\{\Omega_2 - \Omega_1, \Omega_4 - \Omega_3\}. \quad (1.182)$$

Note that the output is composed of one sample of each signal $x_1(n)$ and $x_2(n)$; therefore, the effective sampling frequency is $2\Omega_s$. $\triangle$

As illustrated in the example above, for some bandpass signals $x(t)$, sampling may be performed below the limit $2\Omega_{\max}$, where $\Omega_{\max}$ represents the maximum absolute value of the frequency present in $x(t)$. Although one cannot obtain a general expression for the minimum Ω_s in these cases, it must always satisfy $\Omega_s > \Delta\Omega$, where $\Delta\Omega$ represents the net bandwidth of $x(t)$.

As mentioned in Theorem 1.1, the original continuous-time signal $x_a(t)$ can be recovered from the signal $x_i(t)$ by filtering $x_i(t)$ with an ideal lowpass filter having cutoff frequency $\Omega_s/2$. More specifically, if the signal has bandwidth Ω_c , then it suffices that the cutoff frequency of the ideal lowpass filter is Ω_{LP} , such that $\Omega_c \leq \Omega_{LP} \leq \Omega_s/2$. Therefore, the

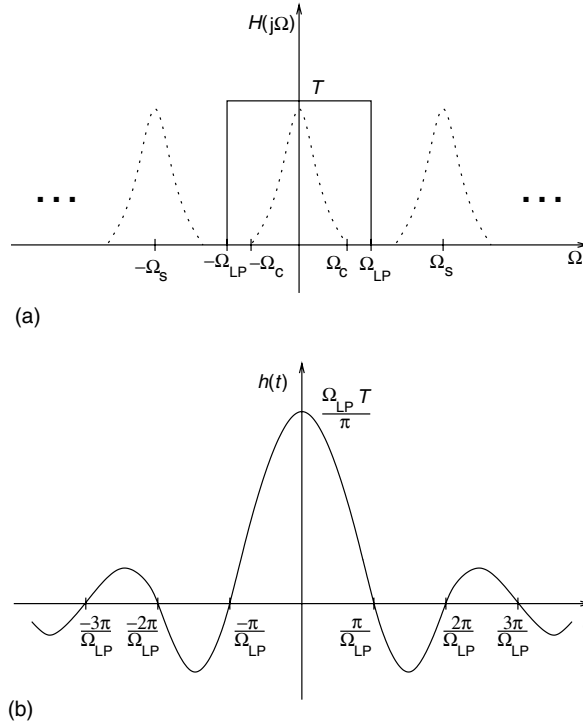


Fig. 1.13. Ideal lowpass filter: (a) frequency response; (b) impulse response.

Fourier transform of the impulse response of such a filter should be

$$H(j\Omega) = \begin{cases} T, & \text{for } |\Omega| < \Omega_{LP} \\ 0, & \text{for } |\Omega| \geq \Omega_{LP}, \end{cases} \quad (1.183)$$

where the passband gain T compensates for the factor $1/T$ in Equation (1.170). This ideal frequency response is illustrated in Figure 1.13a.

Computing the inverse Fourier transform of $H(j\Omega)$ using Equation (1.159), we see that the impulse response $h(t)$ of the filter is

$$h(t) = \frac{T \sin(\Omega_{LP} t)}{\pi t}, \quad (1.184)$$

which is depicted in Figure 1.13b.

Then, given $h(t)$, the signal $x_a(t)$ can be recovered from $x_i(t)$ by the following convolution integral (Oppenheim *et al.*, 1983):

$$x_a(t) = \int_{-\infty}^{\infty} x_i(\tau) h(t - \tau) d\tau. \quad (1.185)$$

Replacing $x_i(t)$ by its definition in Equation (1.163), we have that

$$\begin{aligned}
 x_a(t) &= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x(n) \delta(\tau - nT) h(t - \tau) d\tau \\
 &= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(n) \delta(\tau - nT) h(t - \tau) d\tau \\
 &= \sum_{n=-\infty}^{\infty} x(n) h(t - nT),
 \end{aligned} \tag{1.186}$$

and using $h(t)$ from Equation (1.184):

$$x_a(t) = \sum_{n=-\infty}^{\infty} x(n) \frac{T \sin[\Omega_{LP}(t - nT)]}{\pi(t - nT)} \tag{1.187}$$

If we make Ω_{LP} equal to half the sampling frequency, that is, $\Omega_{LP} = \Omega_s/2 = \pi/T$, then Equation (1.187) becomes

$$x_a(t) = \sum_{n=-\infty}^{\infty} x(n) \frac{\sin\left(\frac{\Omega_s}{2}t - n\pi\right)}{\frac{\Omega_s}{2}t - n\pi} = \sum_{n=-\infty}^{\infty} x(n) \frac{\sin\left[\pi\left(\frac{t}{T} - n\right)\right]}{\pi\left(\frac{t}{T} - n\right)}. \tag{1.188}$$

Equations (1.187) and (1.188) represent interpolation formulas to recover the continuous-time signal $x_a(t)$ from its samples $x(n) = x_a(nT)$. However, since, in order to compute $x_a(t)$ at any time t_0 , all the samples of $x(n)$ have to be known, these interpolation formulas are not practical. This is the same as saying that the lowpass filter with impulse response $h(t)$ is not realizable. This is because it is noncausal and cannot be implemented via any differential equation of finite order. Clearly, the closer the lowpass filter is to the ideal, the smaller is the error in the computation of $x(t)$ using Equation (1.187). In Chapters 5 and 6, methods to approximate such ideal filters will be studied extensively.

From what we have studied in this section, we can develop a block diagram of the several phases constituting the processing of an analog signal using a digital system. Figure 1.14 depicts each step of the procedure.

The first block in the diagram of Figure 1.14 is a lowpass filter, that guarantees the analog signal is band-limited, with bandwidth $\Omega_c < \Omega_s/2$.

The second block is a sample-and-hold system, which samples the signal $x_a(t)$ at times $t = nT$ and holds the value obtained for T seconds; that is, until the value for the next time interval is sampled. More precisely, $x_a^*(t) = x_a(nT)$, for $nT < t < (n+1)T$.

The third block, the encoder, converts each sample value output by the sample and hold, $x_a^*(t)$, for $nT < t < (n+1)T$, to a number $x(n)$. Since this number is input to digital hardware, it must be represented with a finite number of bits. This operation introduces an error in the signal, which gets smaller as the number of bits used in the representation

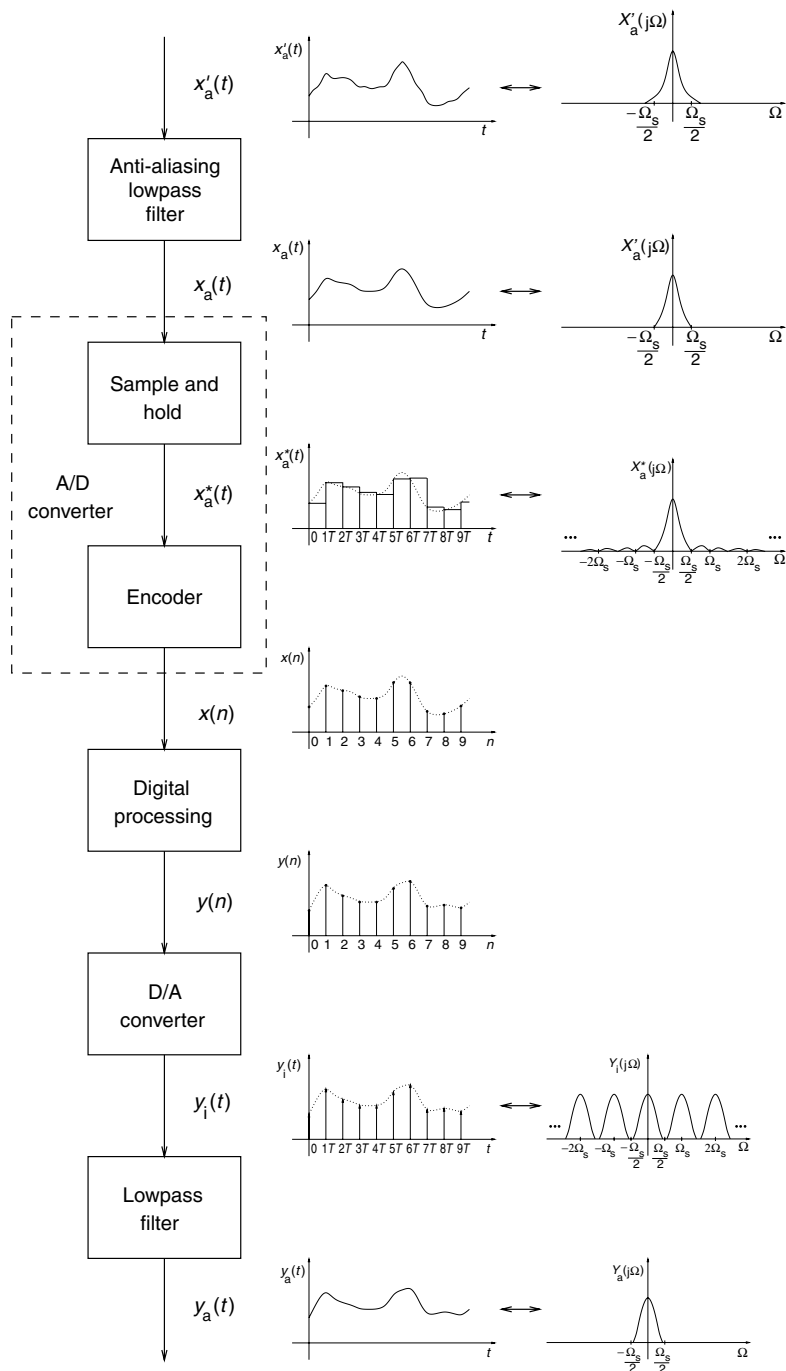


Fig. 1.14.

Digital processing of analog signals (all graphs on the left represent the signals in the time domain and all graphs on the right are their corresponding Fourier transforms).

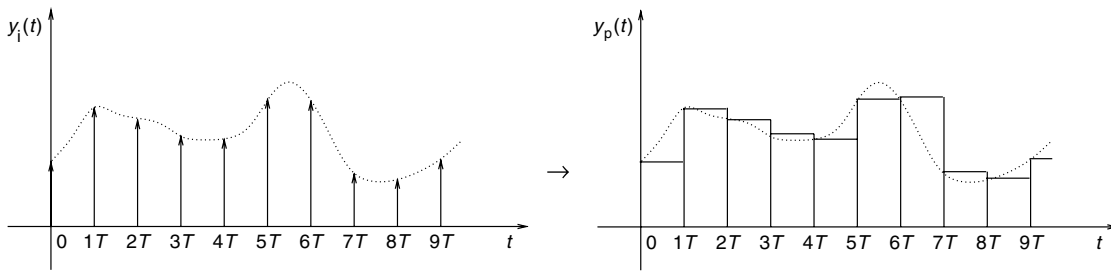


Fig. 1.15. Sample-and-hold operation.

increases. The second and third blocks constitute what we usually call the analog-to-digital (A/D) conversion.

The fourth block carries out the digital signal processing, transforming the discrete-time signal $x(n)$ into a discrete-time signal $y(n)$.

The fifth block transforms the numbers representing the samples $y(n)$ back into the analog domain, in the form of a train of impulses $y_i(t)$, constituting the process known as digital-to-analog (D/A) conversion.

The sixth block is a lowpass filter necessary to eliminate the repetitions of the spectrum contained in $y_i(t)$, in order to recover the analog signal $y_a(t)$ corresponding to $y(n)$. In practice, sometimes the fifth and sixth blocks are implemented in one operation. For example, we can transform the samples $y(n)$ into an analog signal $y_a(t)$ using a D/A converter plus a sample-and-hold operation, similar to the one of the second block. It is easy to show (see Figure 1.15) that the sample-and-hold operation is equivalent to filtering the train of impulses

$$y_i(t) = \sum_{n=-\infty}^{\infty} y(n)\delta(t - nT) \quad (1.189)$$

with a filter having impulse response

$$h(t) = \begin{cases} 1, & \text{for } 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases} \quad (1.190)$$

yielding the analog signal

$$y_p(t) = \sum_{n=-\infty}^{\infty} y(n)h(t - nT). \quad (1.191)$$

In this case, the recovery of the analog signal is not perfect, but is a good enough approximation in some practical cases.

Example 1.14. For the sample-and-hold operation described by Equations (1.189) to (1.191), determine:

- (a) An expression for the Fourier transform of $y_p(t)$ in Equation (1.191) as a function of the Fourier transform of $x_a(t)$ (suppose that $y(n) = x_a(nT)$).

- (b) The frequency response of an ideal lowpass filter that outputs $x_a(t)$ when $y_p(t)$ is applied to its input. Such a filter would compensate for the artifacts introduced by the sample-and-hold operation in a D/A conversion.

Solution

- (a) The train of pulses given by

$$y_p(t) = \sum_{n=-\infty}^{\infty} y(n) h(t - nT) \quad (1.192)$$

is the result of the convolution of an impulse train with the given pulse as follows:

$$y_p(t) = y_i(t) * h(t). \quad (1.193)$$

Using Equation (1.170), in the frequency domain the above equation becomes

$$Y_p(j\Omega) = Y_i(j\Omega)H(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(j\Omega - j\frac{2\pi}{T}k\right)H(j\Omega). \quad (1.194)$$

From Equation (1.190), since

$$\begin{aligned} H(j\Omega) &= \int_{-\infty}^{\infty} h(t) e^{-j\Omega t} dt \\ &= \int_0^T e^{-j\Omega t} dt \\ &= \frac{1}{j\Omega} (1 - e^{-j\Omega T}) \\ &= \frac{e^{-j\Omega T/2}}{j\Omega} (e^{j\Omega T/2} - e^{-j\Omega T/2}) \\ &= \frac{e^{-j\Omega T/2}}{j\Omega} [2j \sin(\Omega T/2)] \\ &= T e^{-j\Omega T/2} \left[\frac{\sin(\Omega T/2)}{\Omega T/2} \right], \end{aligned} \quad (1.195)$$

hence

$$Y_p(j\Omega) = e^{-j\Omega T/2} \left[\frac{\sin(\Omega T/2)}{\Omega T/2} \right] \sum_{k=-\infty}^{\infty} X_a\left(j\Omega - j\frac{2\pi}{T}k\right). \quad (1.196)$$

- (b) In order to recuperate signal $x_a(t)$, we have to compensate the distortion introduced by the frequency spectrum of the pulse $h(t)$. This can be done by processing $y_p(t)$ with a lowpass filter with the following desirable frequency response:

$$G(j\Omega) = \begin{cases} 0, & |\Omega| \geq \pi/T \\ \frac{T}{H(j\Omega)} = e^{j\Omega T/2} \left[\frac{\Omega T/2}{\sin(\Omega T/2)} \right], & |\Omega| < \pi/T. \end{cases} \quad (1.197)$$

△

Example 1.15. Cinema is a way of storing and displaying moving pictures. Before it was invented, all that was known was a way to store and display single images, by taking photographs. Cinema was invented when someone decided to capture a moving image as a series of pictures equally spaced in time. For example, today, in a commercial movie one captures 24 pictures/second. This scheme works because the human visual system enjoys a property called persistence of vision: when a light is flashed, one still sees the light for some time after it is gone. Because of this, when displaying the pictures in sequence, a human viewer has the illusion of continuous movement.

In mathematical terms, a moving picture is a three-dimensional signal, with two continuous spatial dimensions (representing, for instance, horizontal and vertical directions) and one continuous time dimension. A photograph is a time sample of this three-dimensional signal. When one displays this sequence of time samples (photographs), the human visual system sees it as a continuously moving picture; that is, as a continuous-time signal. From what has been said, we can note that cinema can be regarded as a discrete-time signal processing system.

By referring to Figure 1.14, identify in the cinema context which signal processing operation corresponds to each step of the processes of recording and displaying moving pictures, highlighting the associated design constraints.

Solution

The light field of a gray-scale moving picture can be described as a three-dimensional function $f_a(x, y, t)$, where x and y are spatial coordinates and t is a time coordinate, as represented in Figure 1.16.

The light intensity (luminance) of a point of coordinates (x_0, y_0) is thus the one-dimensional time signal

$$g_a(t) = f_a(x_0, y_0, t), \quad (1.198)$$

as depicted in Figure 1.17.

When one takes photographs of a moving picture every T time units, the three-dimensional signal is sampled in time, generating the discrete-time signal

$$f(x, y, n) = f_a(x, y, nT). \quad (1.199)$$

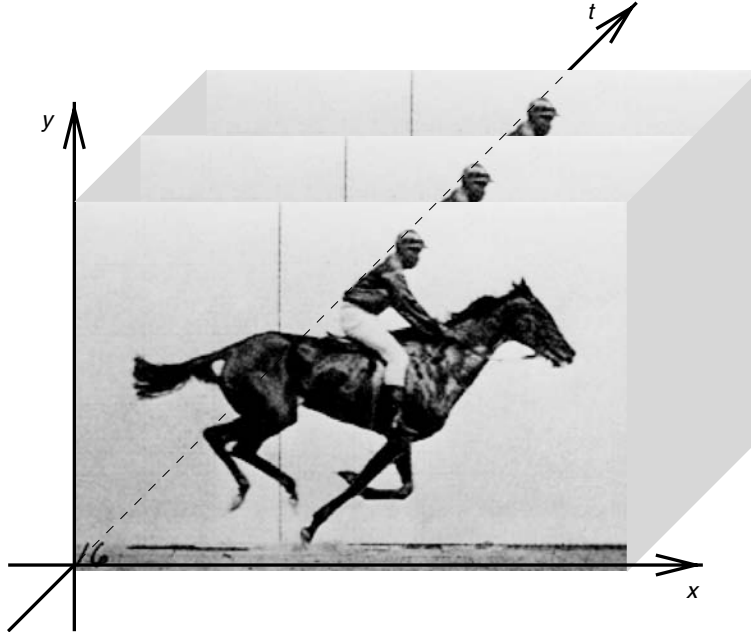


Fig. 1.16. Representation of a moving picture as a three-dimensional signal.

The discrete-time one-dimensional signal corresponding to the intensity of the point (x_0, y_0) is then

$$g(n) = g_a(nT) = f(x_0, y_0, n) = f_a(x_0, y_0, nT), \quad (1.200)$$

as represented in Figure 1.18.

Therefore, in order to avoid aliasing, one should sample the signal $g_a(t)$ with a sampling frequency larger than twice its bandwidth. Supposing that the bandwidth of $g_a(t) = f_a(x_0, y_0, t)$ is W_{x_0, y_0} , then, in order to avoid aliasing, the time interval between photographs should be determined by the largest time bandwidth among all coordinates of the picture; that is:

$$T < \frac{2\pi}{\max_{x_0, y_0} \{2W_{x_0, y_0}\}}. \quad (1.201)$$

It is reasonable to think that the faster the objects in a scene move, the larger is the time bandwidth of the light intensity of its points. Therefore, Equation (1.201) says that, in order to avoid aliasing, one should film the scene by taking enough pictures per second such that the time interval between them satisfies a maximum constraint. For example, when filming a hummingbird, if there is too long an interval between photographs, it might flap its wings many times between photographs. Depending on the speed one takes the photographs, its

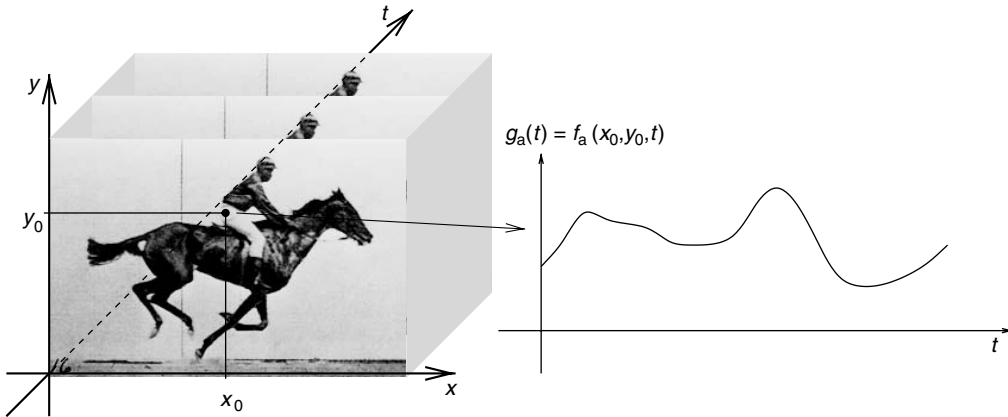


Fig. 1.17. Time signal generated by the intensity of a point (x_0, y_0) from a moving picture.

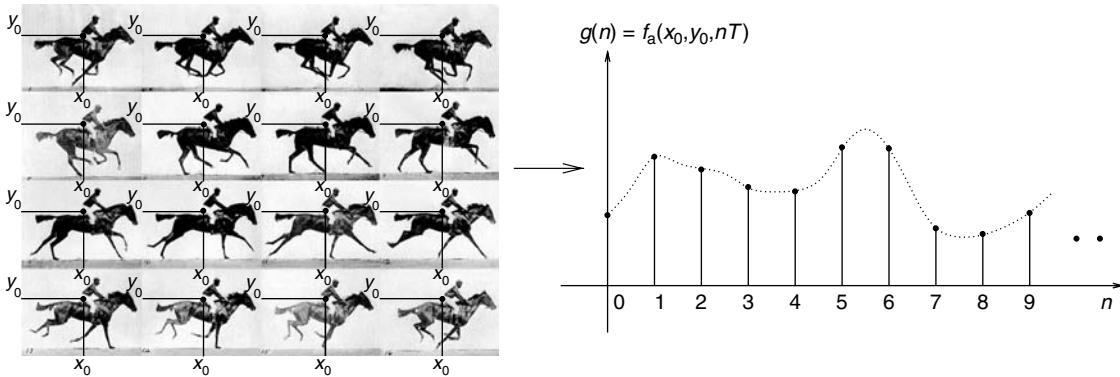


Fig. 1.18. Discrete-time signal generated by the intensity of a given coordinate from photographs of a moving picture.

wings may even be approximately in the same position in every photograph. This would have the effect that its wings would appear static when playing the film. This is a good example of the aliasing that occurs when a three-dimensional space-time signal is inadequately sampled in time.

By referring to Figure 1.14, we can see that the process of filming (that is, taking pictures of a moving scene equally spaced in time) is equivalent to the “Sample and hold” and “Encoder” blocks. It is interesting to note that in this photograph-shooting context there cannot be any anti-aliasing filter. This is so because one cannot change the way a scene varies in time before filming it. Therefore, depending on how fast a scene is moving, aliasing cannot be avoided.

As seen above, through the sampling process, one is able to represent a two-dimensional moving picture (a three-dimensional signal) as a sequence of photographs. These photographs can be stored and processed. After that, one has to display a moving picture from these photographs. In the cinema, each photograph is usually in the form of a transparency. With the help of a system of lenses, one transparency can be projected on a screen by turning on a lamp behind it. In order to display a sequence of transparencies on the screen, there must be a way of replacing the transparency currently in front of the lamp by the next one. It is easier to do this if the lamp is turned off during the transparency replacement process. This is the same as having a flashing light source projecting the transparencies on the screen such that a different transparency is in front of the light source each time it flashes. By modeling the intensity of a flashing light as an impulse, a light flashing periodically can be represented by a train of impulses, allowing one to depict the moving-picture displaying process as given in Figure 1.19. There, the light intensity of a given point on the screen has been modeled as a train of impulses, each impulse having as amplitude the intensity of the

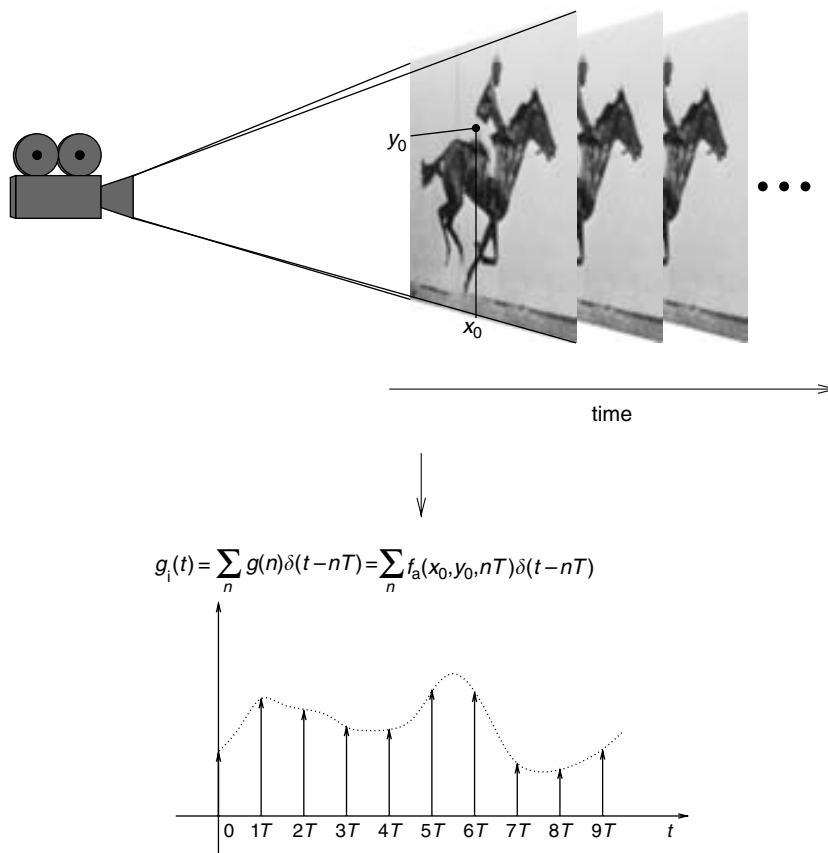


Fig. 1.19.

The film projector is equivalent to replacing the intensity of each sample in a picture by an impulse.

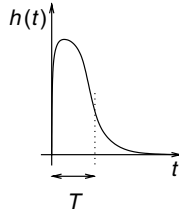


Fig. 1.20.

Response of the human visual system to a flash of light characterizing the persistence of vision phenomenon.

point at a given time. If the light field projected on the screen is $f_i(x, y, t)$, then we have that

$$f_i(x, y, t) = \sum_{n=-\infty}^{\infty} f_a(x, y, nT) \delta(t - nT). \quad (1.202)$$

Thus, using Equation (1.200), the intensity at point (x_0, y_0) is

$$g_i(t) = f_i(x_0, y_0, t) = \sum_{n=-\infty}^{\infty} f_a(x_0, y_0, nT) \delta(t - nT) = \sum_{n=-\infty}^{\infty} g(n) \delta(t - nT), \quad (1.203)$$

which is equivalent to Equations (1.163) and (1.164).

The human visual system enjoys a property called persistence of vision. In a nutshell, if a light is flashed before your eyes, persistence of vision is the reason why you keep seeing the light for some time after it goes off. This is depicted in Figure 1.20, where $h(t)$ represents the human visual system response to a flash of light at $t = 0$.

Owing to the persistence of vision, when you look at a sequence of pictures flashing in the screen, you do not actually see the pictures flashing provided that the flashes are frequent enough. Instead, you have the impression that the picture is moving continuously. In mathematical terms, the function $h(t)$ in Figure 1.20 is the response of the human visual system to an impulse of light $\delta(t)$. Therefore, from Equation (1.203), the response of the human visual system to a point (x_0, y_0) flashing on the cinema screen is given by

$$g_r(t) = g_i(t) * h(t) = \sum_{n=-\infty}^{\infty} g(n) h(t - nT) = \sum_{n=-\infty}^{\infty} f_a(x, y, nT) h(t - nT) \quad (1.204)$$

which is equivalent to Equations (1.185) and (1.186). By referring to Figure 1.21, the human visual system replaces each impulse by the function $h(t)$ in Figure 1.20, thus perceiving the sequence of light impulses as the right-hand-side signal in Figure 1.21.

At this point, we can make interesting design considerations about the cinema system. As we have seen before, the time interval between photographs is limited by the aliasing effect, and is given by Equation (1.201). In a regular signal processing system, one would choose the impulse response of the reconstructing filter according to Equations (1.183) and (1.184), so that the spectrum repetitions are filtered out from the train of impulses and the original analog signal is recovered. However, in cinema, the impulse response

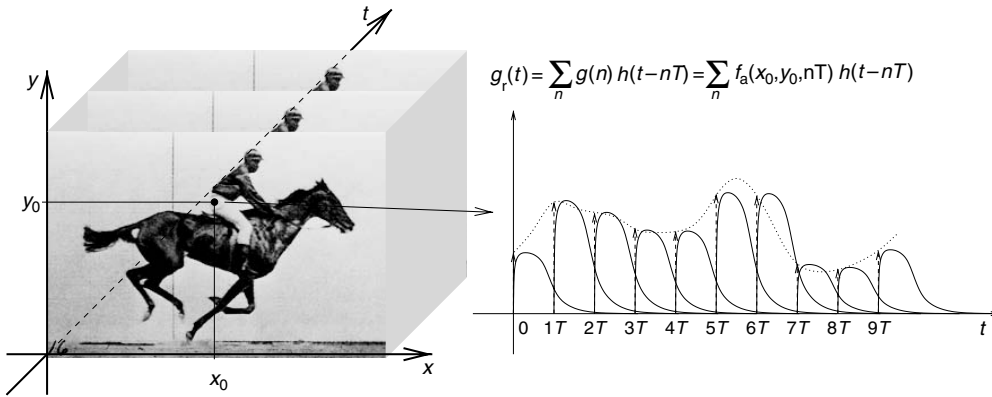


Fig. 1.21. Through persistence of vision, the human visual system replaces each flash of light by a function $h(t)$.

of the reconstructing filter is given by the persistence of vision, which is a physiological characteristic, and thus cannot be changed. Specifically in modern cinema, in order to avoid aliasing in most scenes of interest, it is enough that we take pictures at a rate of 24 pictures per second; that is, $T \leq 1/24$ s. Therefore, according to Equation (1.183), in order to filter out the spectrum repetitions, the bandwidth of the time impulse response of the human visual system $h(t)$ should be $\Omega_{LP} < 24$ Hz. However, psychovisual tests have determined that, for the human visual system, $h(t)$ is a lowpass function with $\Omega_{LP} \approx 48$ Hz. With this natural cutoff frequency, the spectrum repetitions cannot be filtered out, and thus one loses the impression of a continuous movement. In this case, one perceives that the pictures are flashing, a phenomenon referred to as flickering. In order to avoid this, there are two options. One of them is to halve the sampling period, matching T to $\Omega_{LP} = 48$ Hz. This solution has the disadvantage of requiring twice as many pictures as the ones required to avoid aliasing. This is not cost effective, since only 24 pictures per second are enough to avoid aliasing. The solution employed by modern cinema is to repeat each picture twice so that the interval between pictures is $1/48$ s. This procedure avoids aliasing, allowing one to filter out the spectrum repetitions, as determined in Exercise 1.27. The whole movie-generating process is summarized in Figure 1.22. $\triangle$

1.7 Random signals

In nature, we are commonly forced to work with signals whose waveforms are not exactly known at each time instant. Some examples may include the outcome of a die, the suit of a card randomly drawn, a resistor value, or a stock price at a particular time. In such cases, even without knowing the exact signal value, one can still extract valuable information about the process of interest using the mathematical tools presented in this section.

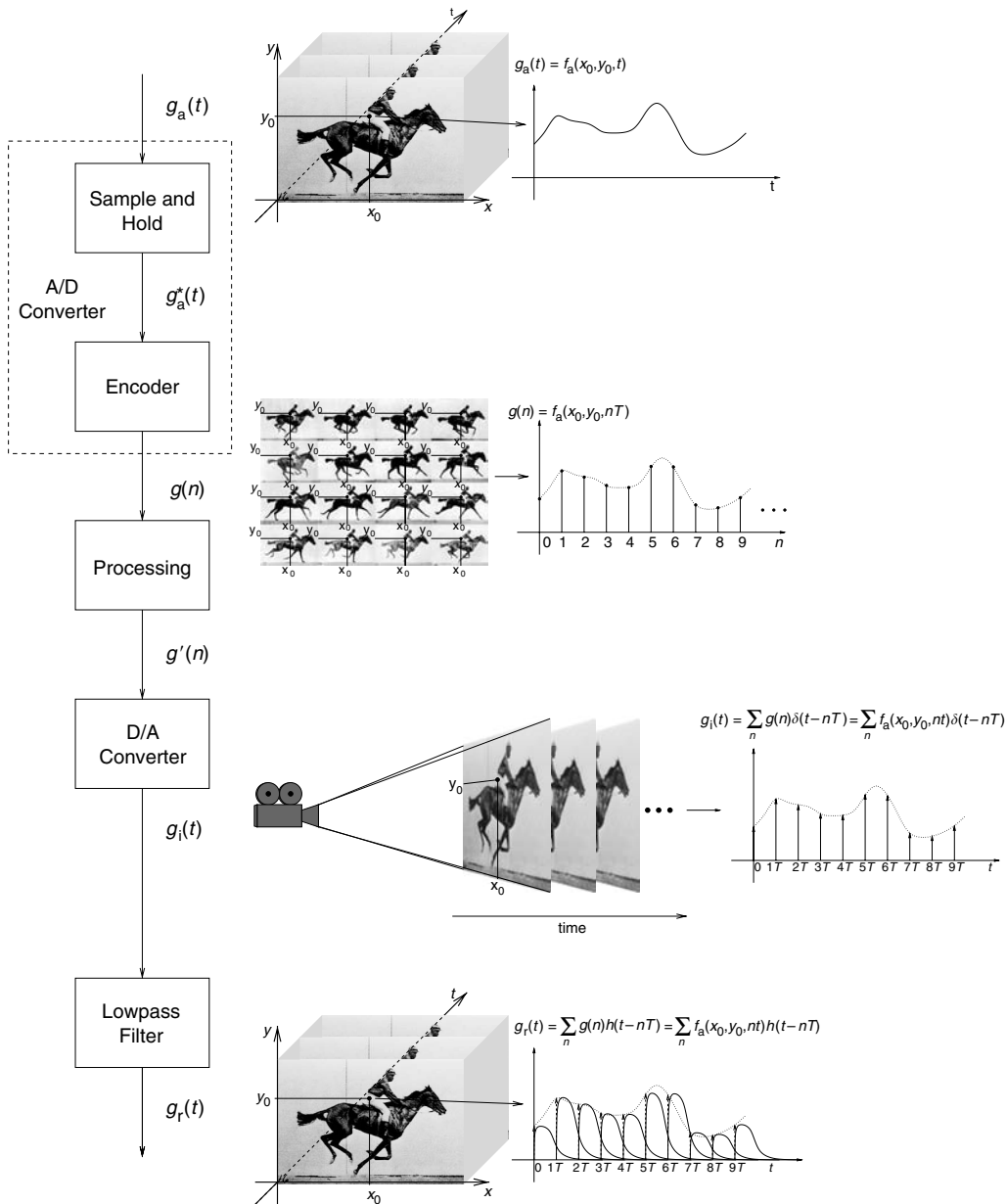


Fig. 1.22. Cinema as a digital signal processing system.

1.7.1 Random variable

A random variable is a mapping of the result of an experiment onto the set of real numbers. By doing so, we can extract numerical information about the problem at hand, as indicated below.

The cumulative distribution function (CDF) is determined by the probability of a given random variable X to be less than or equal to a particular value x ; that is:

$$F_X(x) = P\{X \leq x\}, \quad (1.205)$$

where $P\{\mathcal{E}\}$ denotes the probability of the event $\mathcal{E}$. The corresponding probability density function (PDF) is given by

$$f_X(x) = \frac{dF_X(x)}{dx}. \quad (1.206)$$

From their definitions, it is simple to infer that the CDF and PDF present the following properties:

- $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $\lim_{x \rightarrow +\infty} F_X(x) = 1$
- $F_X(x)$ is a nondecreasing function with x , such that $0 \leq F_X(x) \leq 1$
- $f_X(x) \geq 0$, for all x
- $\int_{-\infty}^{\infty} f_X(x) dx = 1$.

These two functions unify the statistical treatment of both discrete- and continuous-valued random variables, despite the fact that each variable type has probability functions with different characteristics (discrete random variables, for instance, present impulsive PDFs).

There is a plethora of PDFs in the associated literature (Papoulis, 1977; Peebles, 2000) to characterize all kinds of random phenomena. However, for our purposes, the most interesting PDFs are the uniform (continuous $u_{X,c}(x)$ and discrete $u_{X,d}(x)$) and Gaussian $\phi_X(x)$ distributions, defined as

$$u_{X,c}(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise} \end{cases}, \quad (1.207)$$

$$u_{X,d}(x) = \begin{cases} \frac{1}{M} \delta(x - x_i), & x_i = x_1, x_2, \dots, x_M, \\ 0, & \text{otherwise} \end{cases}, \quad (1.208)$$

and

$$\phi_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\bar{\mu})^2/2\sigma^2} \quad (1.209)$$

respectively, where a , b , M , $\bar{\mu}$, and σ^2 are the corresponding parameters for these distributions. Using the PDF, we can extract meaningful measures that characterize the statistical behavior of a given random variable. In particular, we define the i th-order moment as

$$E\{X^i\} = \int_{-\infty}^{\infty} x^i f_X(x) dx, \quad (1.210)$$

where the two most important special cases are the first-order moment (statistical mean or statistical expectation) $\mu_X = E\{X\}$ and the second-order moment (energy or mean-squared value) $E\{X^2\}$. If Y is a function of the random variable X (that is, $Y = g(X)$), then for the first-order moment of Y we can write that

$$\mu_Y = E\{Y\} = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_{-\infty}^{\infty} g(x) f_X(x) dx. \quad (1.211)$$

We are often interested in measures that are not influenced by a random variable's mean value. For these cases, we may also define the so-called i th-order central moments as

$$E\{(X - \mu_X)^i\} = \int_{-\infty}^{\infty} (x - \mu_X)^i f_X(x) dx. \quad (1.212)$$

By far the most important central moment is the second-order one, also known as the variance $\sigma_X^2 = E\{(X - \mu_X)^2\}$, the square-root of which is referred to as the standard deviation. Larger variances indicate that the value of the random variable is more spread around the mean, whereas smaller σ_X^2 occurs when the values of X are more concentrated around its mean. A simple algebraic development indicates that

$$\sigma_X^2 = E\{(X - \mu_X)^2\} = E\{X^2\} - 2E\{X\mu_X\} + \mu_X^2 = E\{X^2\} - \mu_X^2. \quad (1.213)$$

Example 1.16. Determine the values for the statistical mean, energy, and variance of a random variable X characterized by the discrete uniform PDF given in Equation (1.208) with $x_i = 1, 2, \dots, M$.

Solution

Using the corresponding definitions, where the integral operations on the delta functions degenerate into simple sums, we get

$$\mu_X = \sum_{i=1}^M i \frac{1}{M} = \frac{1}{M} \left[\frac{1}{2} M(M+1) \right] = \frac{M+1}{2} \quad (1.214)$$

$$E\{X^2\} = \sum_{i=1}^M i^2 \frac{1}{M} = \frac{1}{M} \left[\frac{M(M+1)(2M+1)}{6} \right] = \frac{2M^2 + 3M + 1}{6}. \quad (1.215)$$

The variance can be determined by the relationship expressed in Equation (1.213), yielding

$$\sigma_X^2 = \frac{M^2 - 1}{12}. \quad (1.216)$$

All statistics presented above are based on the PDF of the random variable. In that sense, we are exchanging the requirement of knowing the value of a variable by the requirement of knowing its PDF. This may seem like trading one problem for another. However, in several cases, even though we are not able to determine the value of X , we can still determine its PDF. One simple example is the outcome of casting a fair die, which cannot be predicted but

whose associated PDF is easily determined. This allows one to know important statistics associated with the event, as illustrated in this example by considering $M = 6$. $\triangle$

When dealing with two random variables X and Y simultaneously, we may define the joint CDF by

$$F_{X,Y}(x,y) = P\{X \leq x, Y \leq y\} \quad (1.217)$$

and the corresponding joint PDF as

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y}. \quad (1.218)$$

The two variables are said to be statistically independent if we may write

$$f_{X,Y}(x,y) = f_X(x)f_Y(y), \quad (1.219)$$

where independence indicates that the outcome of a given variable does not affect the value of the other. The concept of moments is easily extended to two random variables X and Y by defining the joint moment of order (i,j) as

$$E\{X^i Y^j\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^i y^j f_{X,Y}(x,y) \, dx \, dy \quad (1.220)$$

and the joint central moment of order (i,j) as

$$E\{(X - \mu_X)^i (Y - \mu_Y)^j\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)^i (y - \mu_Y)^j f_{X,Y}(x,y) \, dx \, dy. \quad (1.221)$$

When the orders are $(1, 1)$, we get the cross-correlation $r_{X,Y}$ and the cross-covariance $c_{X,Y}$ between X and Y ; that is:

$$\begin{aligned} r_{X,Y} &= E\{XY\} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x,y) \, dx \, dy \end{aligned} \quad (1.222)$$

$$\begin{aligned} c_{X,Y} &= E\{(X - \mu_X)(Y - \mu_Y)\} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_{X,Y}(x,y) \, dx \, dy \end{aligned} \quad (1.223)$$

In the case of complex random variables, the cross-correlation is given by $r_{X,Y} = E\{XY^*\}$, where the superscript asterisk denotes the complex conjugation operation. For the sake of simplicity, the remainder of this chapter is restricted to real signals.

1.7.2 Random processes

A random process is an ordered collection of random variables. The most common form of ordering the set of random variables is associating each of them with different time instants, giving rise to the common interpretation of a random process as a set of random variables ordered in time.

The whole concept can be better understood with the assistance of a practical example. Consider the complete set of utterances of the vowel /A/ by a particular person. We refer to the m th discrete-time utterance as $a_m(n)$, for $m = 1, 2, \dots$. For a given m it can be regarded as a sample or a realization of the random process $\{A\}$. The entire set of realizations is referred to as the ensemble. In the context of signal processing, a realization may also be referred to as a random signal. If we consider a particular time instant n_1 , the value of each random signal at n_1 defines the random variable $A(n_1)$; that is:

$$A(n_1) = \{a_1(n_1), a_2(n_1), \dots\}. \quad (1.224)$$

Naturally, for the whole random process $\{A\}$ we can define an infinite number of random variables $A(n)$, each one associated with a particular time instant n and all of them obeying the intrinsic order established by the time variable.

In a random process $\{X\}$, each random variable $X(n)$ has its own PDF $f_{X(n)}(x)$, which can be used to determine the i th-order moments of the associated random variable. Similarly, two random variables $X(n_1)$ and $X(n_2)$ of the same random process have a joint PDF defined by

$$f_{X(n_1), X(n_2)}(x_1, x_2) = \frac{\partial^2 P\{X(n_1) \leq x_1, X(n_2) \leq x_2\}}{\partial x_1 \partial x_2}. \quad (1.225)$$

Based on this function, we can define the so-called autocorrelation function of the random process,

$$\begin{aligned} R_X(n_1, n_2) &= r_{X(n_1), X(n_2)} \\ &= E\{X(n_1)X(n_2)\} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X(n_1), X(n_2)}(x_1, x_2) dx_1 dx_2. \end{aligned} \quad (1.226)$$

As its name indicates, the autocorrelation function represents the statistical relation between two random variables (associated with the time-instants n_1 and n_2) of a given random process.

A random process is called wide-sense stationary (WSS) if its mean value and autocorrelation function present the following properties:

$$E\{X(n)\} = c, \text{ for all } n \quad (1.227)$$

$$R_X(n, n + v) = R_X(v), \text{ for all } n, v. \quad (1.228)$$

The first relation indicates that the mean value of all random variables $X(n)$ is constant throughout the entire process. The second property means that the autocorrelation function

of a WSS process depends only on the time interval between two random variables, not on their absolute time instants. Overall, the two relationships indicate that the first- and second-order statistics of the random process do not change over time, indicating the (wide-sense) stationary nature of the process from a statistical point of view. If the statistics of all orders are invariant in time, the process is referred to as strict-sense stationary (SSS). Hence, SSS processes are also WSS, whereas the opposite is not necessarily true. Since it is very hard to verify the invariance property for all orders, in practice we often work with the WSS characterization, which only requires first- and second-order invariance. Later, we verify that WSS random processes, for instance, can be well characterized also in the frequency domain. The stationarity concept can be extended to two distinct processes $\{X\}$ and $\{Y\}$, which are said to be jointly WSS if their first-order moments are constant for all n and if their cross-correlation function $R_{XY}(n, n + \nu) = E\{X(n)Y(n + \nu)\}$ is only a function of ν .

Example 1.17. Consider the random process $\{X\}$ described by its m th realization as

$$x_m(n) = \cos(\omega_0 n + \Theta_m), \quad (1.229)$$

where Θ is a continuous random variable with uniform PDF within the interval $[0, 2\pi]$. Determine the statistical mean and the autocorrelation function for this process, verifying whether or not it is WSS.

Solution

By writing that $X(n) = g(\Theta)$, from Equations (1.211) and (1.226) respectively we get

$$\begin{aligned} E\{X(n)\} &= \int_0^{2\pi} g(\theta) f_{\Theta}(\theta) d\theta \\ &= \int_0^{2\pi} \cos(\omega_0 n + \theta) \frac{1}{2\pi} d\theta \\ &= \frac{1}{2\pi} \sin(\omega_0 n + \theta) \Big|_{\theta=0}^{\theta=2\pi} \\ &= \frac{1}{2\pi} [\sin(\omega_0 n + 2\pi) - \sin(\omega_0 n)] \\ &= 0, \text{ for all } n \end{aligned} \quad (1.230)$$

$$\begin{aligned} R_X(n_1, n_2) &= E\{\cos(\omega_0 n_1 + \Theta) \cos(\omega_0 n_2 + \Theta)\} \\ &= \frac{1}{2} E\{\cos(\omega_0 n_1 - \omega_0 n_2)\} + \frac{1}{2} E\{\cos(\omega_0 n_1 + \omega_0 n_2 + 2\Theta)\}. \end{aligned} \quad (1.231)$$

In this last development for $R_X(n_1, n_2)$, the first term is not random and the corresponding expected value operator can be dropped, whereas for the second term one has that

$$\begin{aligned} \frac{1}{2} E\{\cos(\omega_0 n_1 + \omega_0 n_2 + 2\Theta)\} &= \frac{1}{2} \int_0^{2\pi} \cos(\omega_0 n_1 + \omega_0 n_2 + 2\theta) \frac{1}{2\pi} d\theta \\ &= \frac{1}{8\pi} \sin(\omega_0 n_1 + \omega_0 n_2 + 2\theta) \Big|_{\theta=0}^{\theta=2\pi} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{8\pi} [\sin(\omega_0 n_1 + \omega_0 n_2 + 4\pi) - \sin(\omega_0 n_1 + \omega_0 n_2)] \\
&= 0
\end{aligned} \tag{1.232}$$

and thus

$$R_X(n_1, n_2) = \frac{1}{2} \cos(\omega_0 n_1 - \omega_0 n_2) = \frac{1}{2} \cos[\omega_0(n_1 - n_2)], \text{ for all } n_1, n_2. \tag{1.233}$$

Therefore, from Equations (1.230) and (1.233), we conclude that the random process $\{X\}$ is WSS. $\triangle$

The N th-order autocorrelation matrix $\mathbf{R}_X$ of a WSS random process $\{X\}$ is defined as

$$\mathbf{R}_X = \begin{bmatrix} R_X(0) & R_X(1) & R_X(2) & \cdots & R_X(N-1) \\ R_X(-1) & R_X(0) & R_X(1) & \cdots & R_X(N-2) \\ R_X(-2) & R_X(-1) & R_X(0) & \cdots & R_X(N-3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ R_X(1-N) & R_X(2-N) & R_X(3-N) & \cdots & R_X(0) \end{bmatrix} \tag{1.234}$$

with $R_X(v) = E\{X(n)X(n+v)\}$ as before.

We classify a random process as ergodic if all statistics of the ensemble can be determined by averaging on a single realization across the different samples. For example, when the discrete variable corresponds to time, time averaging is performed in order to determine the statistics. Despite being a quite strong assumption, ergodicity is commonly resorted to in situations where only a few realizations or possibly one realization of the random process is available. In such cases, we may drop the realization index m , denoting the available random signal by just $x(n)$. The mean value, variance, autocorrelation function, and so on of the entire process $\{X\}$ are estimated based solely on $x(n)$.

1.7.3 Filtering a random signal

Consider the input–output relationship of a linear time-invariant system, described by its impulse response $h(n)$, given by the convolution sum

$$y(n) = \sum_{k=-\infty}^{\infty} x(n-k)h(k). \tag{1.235}$$

If $x(n)$ is a random signal, the nature of the output signal shall also be random. Let us proceed to characterize the output signal $y(n)$ when $x(n)$ is a WSS random signal.

Determining the mean value of $y(n)$, one gets

$$\begin{aligned}
 E\{y(n)\} &= E \left\{ \sum_{k=-\infty}^{\infty} x(n-k)h(k) \right\} \\
 &= \sum_{k=-\infty}^{\infty} E\{x(n-k)\}h(k) \\
 &= E\{x(n)\} \sum_{k=-\infty}^{\infty} h(k),
 \end{aligned} \tag{1.236}$$

where we have used the facts that $h(n)$ is a deterministic signal and that $E\{x(n)\}$ is constant for all n , since $\{X\}$ is assumed to be WSS. An interesting interpretation of the above equation comes from the fact that the output $w(n)$ of a system to a constant input $v(n) = c$ is

$$w(n) = \sum_{k=-\infty}^{\infty} h(k)v(n-k) = c \sum_{k=-\infty}^{\infty} h(k). \tag{1.237}$$

This means that the output of a linear system to a constant input is also constant, and equal to the input multiplied by the constant

$$H_0 = \sum_{k=-\infty}^{\infty} h(k), \tag{1.238}$$

which can be regarded as the system DC gain. Hence, Equation (1.236) indicates that the statistical mean of the output signal is the mean value of the WSS input signal multiplied by the system DC gain, which is quite an intuitive result.

The autocorrelation function of the output signal is given by

$$\begin{aligned}
 R_Y(n_1, n_2) &= E\{y(n_1)y(n_2)\} \\
 &= E \left\{ \left(\sum_{k_1=-\infty}^{\infty} x(n_1 - k_1)h(k_1) \right) \left(\sum_{k_2=-\infty}^{\infty} x(n_2 - k_2)h(k_2) \right) \right\} \\
 &= E \left\{ \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} x(n_1 - k_1)x(n_2 - k_2)h(k_1)h(k_2) \right\} \\
 &= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} E\{x(n_1 - k_1)x(n_2 - k_2)\}h(k_1)h(k_2) \\
 &= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} R_X(n_1 - k_1, n_2 - k_2)h(k_1)h(k_2).
 \end{aligned} \tag{1.239}$$

Since $\{X\}$ is WSS, then one may write that

$$R_Y(n_1, n_2) = \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} R_X((n_1 - n_2) - k_1 + k_2) h(k_1) h(k_2), \quad (1.240)$$

which is also a function of $v = (n_1 - n_2)$ for all time instants n_1, n_2 (see Exercise 1.31). From Equations (1.236) and (1.240), one concludes that if the input $\{X\}$ is WSS, then the output process $\{Y\}$ to a linear time-invariant system is also WSS.

The scope of random variables and processes is quite vast, and this section can only present the tip of the iceberg on these matters. The main aspect that the reader should keep in mind is that even though it may not be feasible to determine the exact value of a given signal, one can still determine its PDF or autocorrelation function, and extract many pieces of information (statistical mean, variance, stationarity behavior, and so on) about it using the methodology introduced here. In several cases, the results from these analyses are all that one needs to describe a given signal or process, as will be verified in several parts of this book.

1.8 Do-it-yourself: discrete-time signals and systems

In this section, we explore the realm of discrete-time signals using MATLAB. You are encouraged to try out the experiments described below, taking note of everything you learn. Extending the experiments, following one's own curiosity, is highly desirable, since there is no better way to learn signal processing other than by doing signal processing. And that is exactly what MATLAB allows us to do in a straightforward manner.

Experiment 1.1

In Example 1.7 we were able to determine a closed-form expression for the impulse response associated with the difference Equation (1.87). A numerical solution for this problem can be determined, for $\alpha = 1.15$ and $0 \leq n \leq 30$, using the following MATLAB commands:

```
alpha = 1.15; N = 30;
x = [1 zeros(1,N)];
y = filter(1, [1 -1/alpha], x);
stem(y);
```

This yields the plot seen in Figure 1.23.

In general, the MATLAB command

```
y = filter([b_0 b_1 ... b_M], [1 a_1 ... a_N], x, Z_i);
```

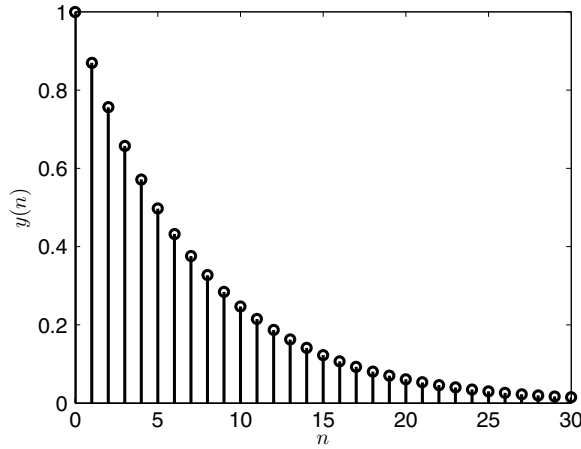


Fig. 1.23. Solution of difference Equation (1.87) in Example 1.7.

determines the solution of the general difference equation

$$y(n) + a_1 y(n-1) + \dots + a_N y(n-N) = b_0 x(n) + b_1 x(n-1) + \dots + b_M x(n-M) \quad (1.241)$$

when the input signal is provided in the vector $\mathbf{x}$ and the vector $\mathbf{Z}_i$ contains its initial conditions.

Experiment 1.2

The process of sampling can be seen as a mapping of a continuous-time function into a set of discrete-time samples. In general, however, there are infinite functions that can generate the same set of samples. To illustrate such a notion, consider a general function $f_1(t)$. Using a sampling frequency of f_s samples per second, the sampling process yields the discrete-time function $f_1(nT_s)$, with $T_s = 1/f_s$ and integer n .

Sampling any function of the form $f_2(t) = f_1(\alpha t)$, with any positive α , using a sampling frequency $f'_s = \alpha f_s$, we get $T'_s = 1/f'_s = T_s/\alpha$. Hence, $f_2(nT'_s) = f_1(\alpha(nT'_s)) = f_1(nT_s)$, which corresponds to the same set of samples as before.

Therefore, in general, a given set of samples does not specify the original continuous-time function in a unique way. To reduce this uncertainty, we must specify the sampling frequency employed to generate the given samples. By doing so, the algebraic reasoning above breaks down and we eliminate (almost) all continuous-time candidate functions for a given sample set. There is, however, one last candidate that must be eliminated to avoid ambiguity. Let us illustrate such case by emulating a sampling procedure using MATLAB.

Consider the 3-Hz cosine function $f_1(t) = \cos(2\pi 3t)$ sampled at $F_s = 10$ samples per second, for a 1s time interval, using the MATLAB command:

```
time = 0:0.1:0.9;
f_1 = cos(2*pi*3.*time)
```

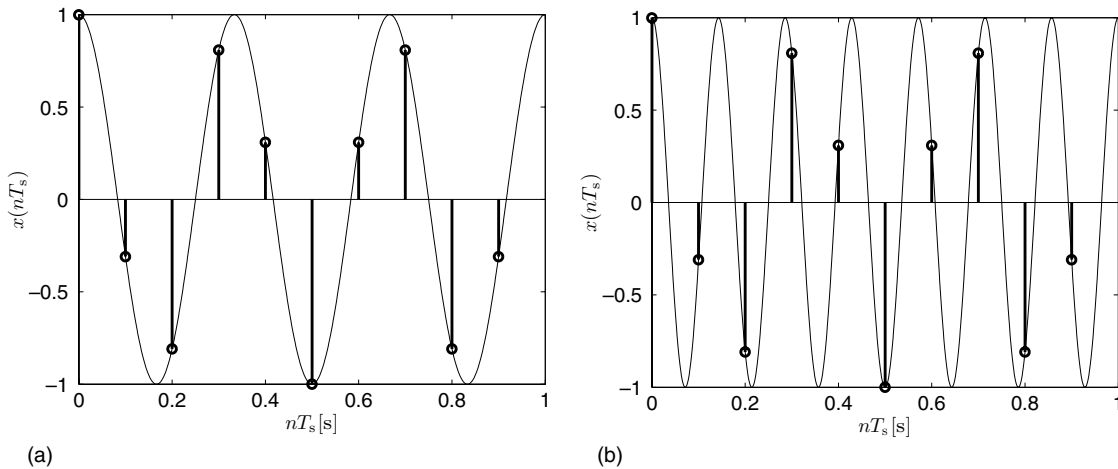


Fig. 1.24. Sampling of cosine functions of frequency f using $F_s = 10$ samples per second: (a) $f = 3$ Hz; (b) $f = 7$ Hz.

An identical sample list can be obtained, with the same sampling rate and time interval, from a 7-Hz cosine function $f_2(t) = \cos(2\pi 7t)$ as given by

$$f_2 = \cos(2\pi \cdot 7 \cdot \text{time})$$

with the variable `time` specified as above. The resulting samples are shown in Figure 1.24, generated by the following commands:

```
time_aux = 0:0.001:(1-0.001);
figure(1);
stem(time, f_1);
hold on;
plot(time_aux, cos(2*pi*3.*time_aux));
hold off;
figure(2);
stem(time, f_2);
hold on;
plot(time_aux, cos(2*pi*7.*time_aux));
hold off;
```

In this sequence, the `hold on` commands allow plotting more than one function in the same figure and the `time_aux` variable is used to emulate a continuous time counter in the `plot` commands to draw the background functions.

To eliminate the ambiguity illustrated in Figure 1.24, we must refer to the sampling theorem introduced in Section 1.6.2. That result indicates that a 7-Hz cosine function shall not be sampled with $F_s = 10$ Hz, since the minimum sampling frequency in this case should be above $F_s = 14$ Hz. Therefore, if Nyquist's sampling criterion is satisfied, there is only one continuous-time function associated with a given set of discrete-time samples and a particular sampling frequency.

Experiment 1.3

Suppose the signal $x(t) = 5 \cos(2\pi 5t) + 2 \cos(2\pi 50t)$, sampled with $F_s = 1000$ samples per second, as shown in Figure 1.25a, is corrupted by a small amount of noise, forming the signal shown in Figure 1.25b generated by the following commands:

```
amplitude_1 = 5; freq_1 = 5;
amplitude_2 = 2; freq_2 = 50;
F_s = 1000; time = 0:1/F_s:(1-1/F_s);
sine_1 = amplitude_1*sin(2*pi*freq_1.*time);
sine_2 = amplitude_2*sin(2*pi*freq_2.*time);
noise = randn(1,length(time));
x_clean = sine_1 + sine_2;
x_noisy = x_clean + noise;
figure(1);
plot(time,x_clean);
figure(2);
plot(time,x_noisy);
```

In particular, the `randn` command generates the specified number of samples of a pseudo-random signal with Gaussian distribution with zero mean and unit variance.

We can minimize the noisy effect by averaging N successive samples of $x(n) = x_{\text{noisy}}$, implementing the following difference equation:

$$y(n) = \frac{x(n) + x(n-1) + \dots + x(n-N+1)}{N}. \quad (1.242)$$

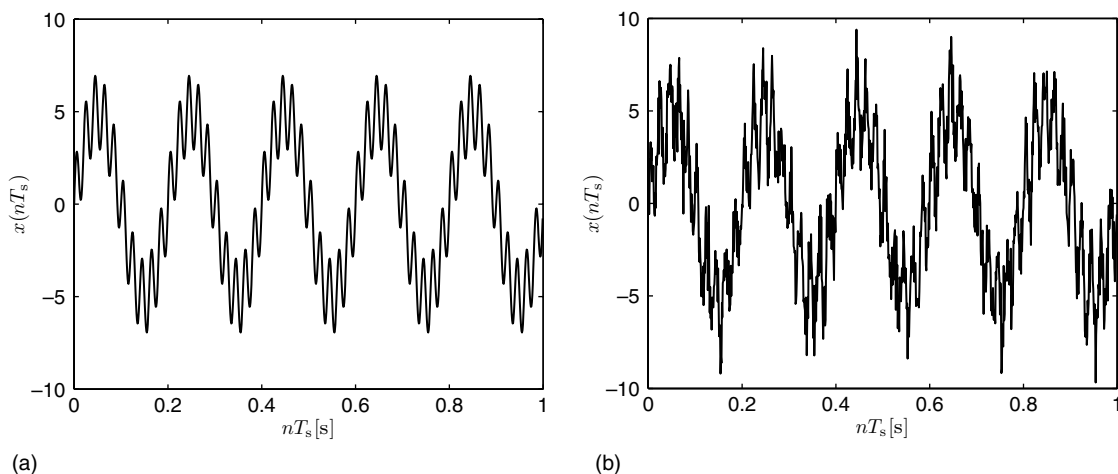


Fig. 1.25. Sum of two sinusoidal components: (a) clean signal; (b) noisy signal.

As mentioned in Experiment 1.1, we can perform such processing by specifying the value of N and using the MATLAB commands

```
b = ones(1,N);  
y = filter(b,1,x_noisy);
```

which yield the plots shown in Figure 1.26 for $N = 3$, $N = 6$, $N = 10$, and $N = 20$.

Figure 1.26 indicates that the averaging technique is quite effective at reducing the amount of noise from the corrupted signal. In this case, the larger the value of N , the greater the ability to remove the noise component. If, however, N is too large, as observed in Figure 1.26d, then the averaging procedure almost eliminates the high-frequency sinusoidal component.

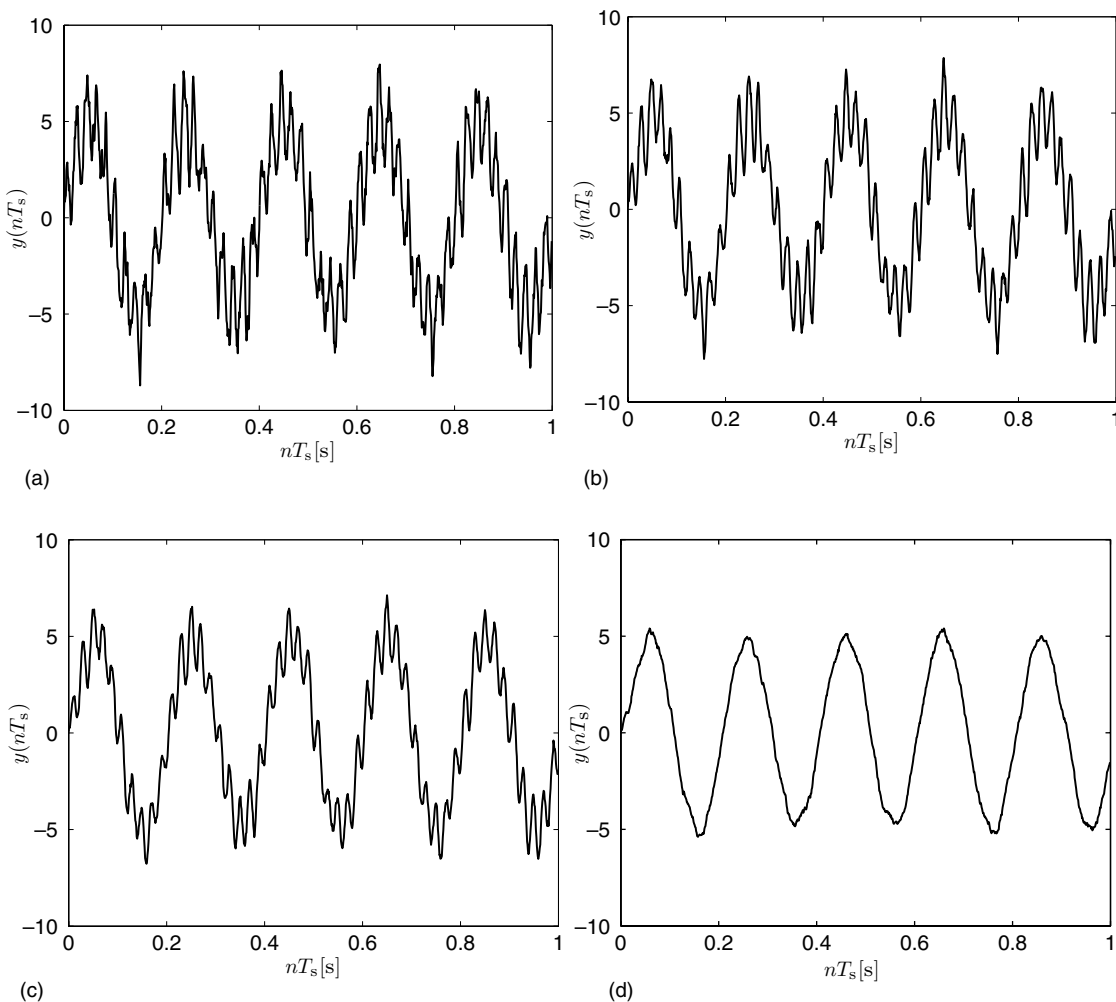


Fig. 1.26. Averaged-in-time signals using N consecutive samples: (a) $N = 3$; (b) $N = 6$; (c) $N = 10$; (d) $N = 20$.

One may then argue whether it is possible to reduce the noise component without affecting the original signal significantly. Perhaps a more elaborate processing may preserve the sinusoidal components better. The theory and design of tools for answering these types of question is the main subject of this textbook. In the following chapters, several techniques for processing a wide range of signals are investigated. Although intuition is important in some practical situations, as illustrated in this experiment, our presentation follows a formal and technically justified path. In the end, the reader may be able not only to employ the methods presented throughout the book, but also to understand them, selecting the proper tool for each particular application.

1.9 Discrete-time signals and systems with MATLAB

The Signal Processing toolbox of MATLAB has several functions that aid in the processing of discrete signals. In this section we give a brief overview of the ones that are most closely related to the material in this chapter.

- **stem**: Plots a data sequence as stems from the x axis terminated in circles, as in the plots of discrete-time signals shown in this chapter.

Input parameters:

- the ordinate values t at which the data is to be plotted;
- the data values x to be plotted.

Example:

```
t=[0:0.1:2]; x=cos(pi*t+0.6); stem(t,x);
```

- **conv**: Performs the discrete convolution of two sequences.

Input parameters: the vectors a and b holding the two sequences.

Output parameter: the vector y holding the convolution of a and b .

Example:

```
a=[1 1 1 1 1]; b=[1 2 3 4 5 6 7 8 9];
c=conv(a,b); stem(c);
```

- **impz**: Computes the impulse response of an initially relaxed system described by a difference equation.

Input parameters (refer to Equation (1.63)):

- a vector b containing the values of b_l , for $l = 0, 1, \dots, M$;
- a vector a containing the values of a_i , for $i = 0, 1, \dots, N$;
- the desired number of samples n of the impulse response.

Output parameter: a vector h holding the impulse response of the system.

Example:

```
a=[1 -0.22 -0.21 0.017 0.01]; b=[1 2 1];
h=impz(b,a,20); stem(h);
```

1.10 Summary

In this chapter we have seen the basic concepts referring to discrete-time systems and have introduced the properties of linearity, time invariance, and causality. Systems presenting these properties are of particular interest due to the numerous tools that are available to analyze and to design them. We also discussed the concept of stability. Linear time-invariant systems have been characterized using convolution sums. We have also introduced difference equations as another way to characterize discrete-time systems. The conditions necessary to carry out the discrete-time processing of continuous-time signals without losing the information they carry have been studied. Random signals have also been introduced. Finally, we presented some hands-on experiments and some MATLAB functions that are useful in the processing of discrete-time signals seen in this chapter.

1.11 Exercises

- 1.1 Characterize the systems below as linear/nonlinear, causal/noncausal and time invariant/time varying:
- (a) $y(n) = (n + a)^2 x(n + 4)$
 - (b) $y(n) = ax(n + 1)$
 - (c) $y(n) = x(n + 1) + x^3(n - 1)$
 - (d) $y(n) = x(n) \sin(\omega n)$
 - (e) $y(n) = x(n) + \sin(\omega n)$
 - (f) $y(n) = \frac{x(n)}{x(n+3)}$
 - (g) $y(n) = y(n - 1) + 8x(n - 3)$
 - (h) $y(n) = 2ny(n - 1) + 3x(n - 5)$
 - (i) $y(n) = n^2 y(n + 1) + 5x(n - 2) + x(n - 4)$
 - (j) $y(n) = y(n - 1) + x(n + 5) + x(n - 5)$
 - (k) $y(n) = (2u(n - 3) - 1)y(n - 1) + x(n) + x(n - 1)$.
- 1.2 For each of the discrete signals below, determine whether they are periodic or not. Calculate the periods of those that are periodic.
- (a) $x(n) = \cos^2\left(\frac{2\pi}{15}n\right)$
 - (b) $x(n) = \cos\left(\frac{4\pi}{5}n + \frac{\pi}{4}\right)$
 - (c) $x(n) = \cos\left(\frac{\pi}{27}n + 31\right)$
 - (d) $x(n) = \sin(100n)$
 - (e) $x(n) = \cos\left(\frac{11\pi}{12}n\right)$
 - (f) $x(n) = \sin[(5\pi + 1)n]$.
- 1.3 Consider the system whose output $y(m)$ is described as a function of the input $x(m)$ by the following difference equations:
- (a) $y(m) = \sum_{n=-\infty}^{\infty} x(n)\delta(m - nN)$
 - (b) $y(m) = x(m) \sum_{n=-\infty}^{\infty} \delta(m - nN)$.
- Determine whether the systems are linear and/or time invariant.